

ETSI EN 300 386 V2.2.1:2022

EN 55032:2015/A11:2020

EN 55035:2017/A11:2020

EN IEC 61000-3-11:2019

EN 61000-3-12:2011

TEST REPORT

For

SWITCHING POWER SUPPLY

MODEL NUMBER: MR484000SG1XXXXXX, MR484000SG2XXXXXX, **MR483000SG3-EF,
MPS484000HL** (X can be A to Z, a to z, 0-9, "-" or blank, indicating the customer code,
which do not affect safety and EMC)

REPORT NUMBER: 4791216025.3

ISSUE DATE: May 29, 2024

Prepared for

**SHENZHEN MEGMEET ELECTRICAL CO LTD
5TH FL, TOWER A B C501-503 D E UNIS INFORPORT, 13 LANGSHAN RD N SEC HI-TECH
INDUSTRIAL PARK NAN SHAN SHENZHEN, GUANGDONG 518057 CHINA**

Prepared by

UL Verification Services (Guangzhou) Co., Ltd, Song Shan Lake Branch

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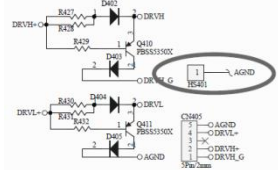
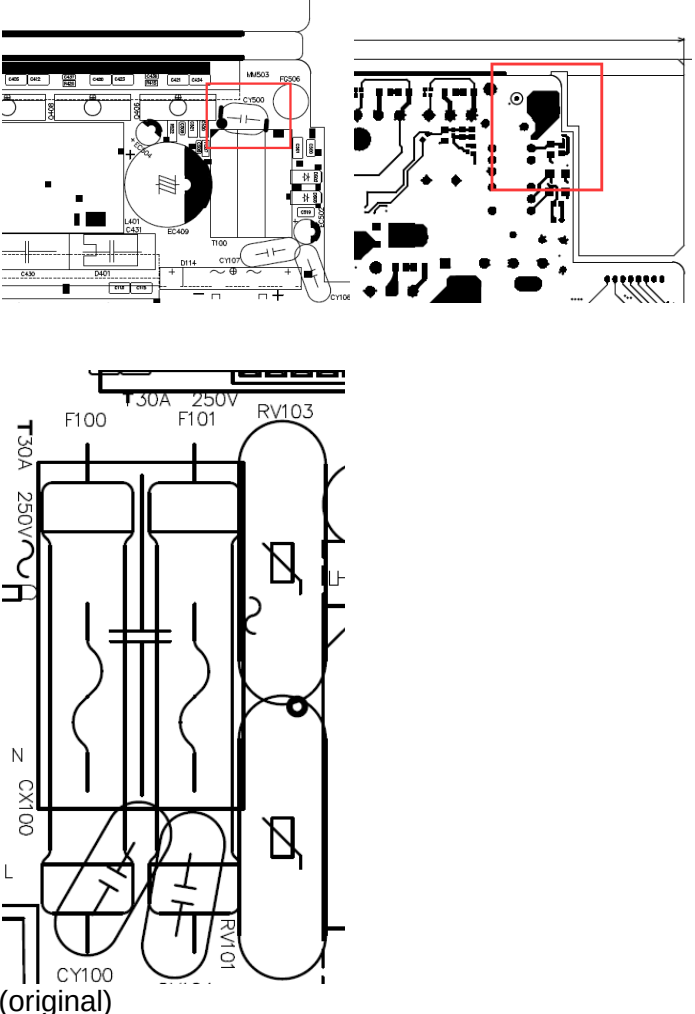
Revision History

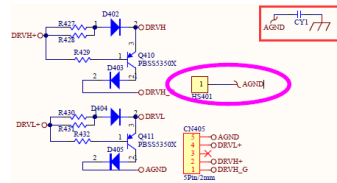
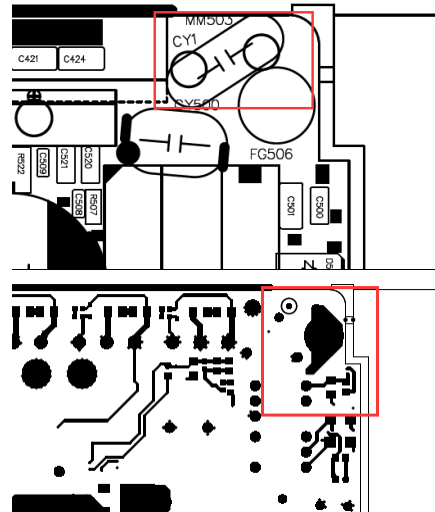
Rev.	Issue Date	Revisions	Revised By
V0	March 22, 2024	Initial Issue	
V1	May 29, 2024	Update	

Note 1: This is a change report based on 4790301933.2-1 issued by UL Certification Services (Guangzhou) Co., Ltd. Songshan Lake Branch on March 23, 2022

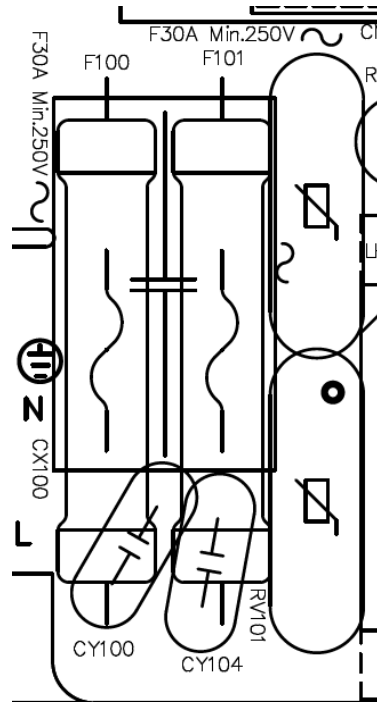
Only for below technical amendment:

- Added a new factory as below:
Altatronic International Co., Ltd.
7/375, 378 Moo 6, Tambon Mabyangporn, Pluak daeng, Rayong, 21140 Thailand
- Revised the model name “MR483000SG3XXXXXX” to “MR483000SG3-EF”.
- Updated the description of variable “X” for below models :
MR484000SG1XXXXXX, MR484000SG2XXXXXX, MR483000SG3-EF, MPS484000HL (X can be A to Z, a to z, 0-9, “-” or blank, indicating the customer code, which do not affect safety and EMC)
- Added a new PCB and circuit due to adding component CY1 for models
MR484000SG2XXXXXX, see below difference and table 4.1.2 for details.

Model	Circuit	PCB layout
MR484000SG2XXXXXX	 (original)	 (original)



(newly added)



(newly added)

Summary of Test Results

Emission				
Standard	Test Item	Limit	Result	
ETSI EN 300 386 V2.2.1:2022 EN 55032:2015/ A11:2020	Conducted emissions (AC mains power ports)	Class B	Pass	
	Radiated emissions below 1GHz	Class B	Pass	
	Radiated emissions above 1GHz	-----	N/A (NOTE 1, 2)	
EN IEC 61000-3- 11:2019	Voltage fluctuations and flicker	-----	Pass	
EN 61000-3-12:2011	Harmonic current emissions	-----	Pass	
Immunity (ETSI EN 300 386 V2.2.1:2022 and EN 55035:2017/A11:2020)				
Basic Standard	Test Item	Test Specification	Criteria	Result
IEC 61000-4-2:2009	Electrostatic Discharge	Contact: +/- 4 kV; +/- 6kV; Air: ± 2 kV; ± 4 kV; ± 8 kV	B	Pass
IEC 61000-4-3:2020	Continuous RF electromagnetic field disturbances	80 MHz-1000 MHz, 1800 MHz,2600 MHz, 3500 MHz,5000 MHz, 3 V/m, 80%, 1 kHz, AM 80 MHz-690 MHz; 3 V/m, 80%, 1 kHz, AM 690-6000 MHz, 10 V/m, 80%, 1 kHz, AM	A	Pass
IEC 61000-4-4:2012	Electrical fast transients burst (AC mains power ports)	+/- 1.0 kV 5/50 ns, 5 kHz	B	Pass
IEC 61000-4- 5:2014+A1 : 2017	Surges (AC mains power ports)	+/-2 kV (Common) +/-1 kV (Differential) 1.2/50 us	B	Pass
IEC 61000-4-6:2014	Continuous induced RF disturbances (AC mains power ports)	80 %, 1 kHz, AM 0.15 MHz-10 MHz: 3 V 10 MHz-30 MHz: 3 V~1 V 30 MHz-80 MHz: 1 V 0.15 MHz-80 MHz: 3 V	A	Pass
IEC 61000-4-8:2009	Power frequency magnetic field	50 Hz, 1 A/m	A	Pass
IEC 61000-4-11:2020	Voltage dips and interruptions (AC mains power ports)	Residual < 5 %: 0.5 cycle; Residual 70 %: 25/30 cycles; Residual < 5 %: 250/300 cycles; Residual voltage 0 %: 1cycle Residual voltage 70 %: 25 cycles Residual voltage 0 %: 250 cycles	B/C/C	Pass

Note:

- 1) N/A: In this whole report not applicable.
- 2) If the highest frequency of the internal sources of the EUT is less than 108 MHz, the measurement shall only be made up to 1 GHz.
If the highest frequency of the internal sources of the EUT is between 108 MHz and 500 MHz, the measurement shall only be made up to 2 GHz.
If the highest frequency of the internal sources of the EUT is between 500 MHz and 1 GHz, the measurement shall only be made up to 5 GHz.
If the highest frequency of the internal sources of the EUT is above 1 GHz, the measurement shall be made up to 5 times the highest frequency or 6 GHz, whichever is less.
- 3) If EUT rated power is less than 75 W, there is no need for Harmonics test to be performed.
- 4) Only applicable to EUT containing devices susceptible to magnetic fields, such as CRT monitors, Hall elements, electrodynamic microphones, magnetic field sensors.
- 5) This test report is only published to and used by the applicant, and it is not for evidence purpose in China.

The measurement result for the sample received is <Pass> according to < ETSI EN 300 386 V2.2.1:2022, EN 55032:2015/A1:2020, EN IEC 61000-3-2:2019/A1:2021, EN 61000-3-3:2013 /A2:2021, EN 55035:2017/A11:2020> when <Accuracy Method> decision rule is applied.

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1. ATTESTATION OF TEST RESULTS

Applicant Information

Company Name: SHENZHEN MEGMEET ELECTRICAL CO LTD
Address: 5th Floor, Tower A, B, C501-C503, D, E Unis Inforport, No.13
Langshan Road North Section, Hi-Tech Industrial Park Nanshan
District 518057 Shenzhen P.R.C

Manufacturer Information

Company Name: SHENZHEN MEGMEET ELECTRICAL CO LTD
Address: 5th Floor, Tower A, B, C501-C503, D, E Unis Inforport, No.13
Langshan Road North Section, Hi-Tech Industrial Park Nanshan
District 518057 Shenzhen P.R.C

Factory Information

Company Name: 1)ZHUZHOU MEGMEET ELECTRICAL CO., LTD
2)Altatronic International Co., Ltd.
Address: 1)1728# TAISHAN RD, TIANYUAN DISTRICT, 412007
ZHUZHOU, HU'NAN, PEOPLE'S REPUBLIC OF CHINA
2)7/375, 378 Moo 6, Tambon Mabyangporn, Pluak daeng,
Rayong, 21140 Thailand

EUT Information

EUT Name: SWITCHING POWER SUPPLY
Model: MR484000SG2-GS
Series Model: MR484000SG1XXXXXX, MR484000SG2XXXXXX,
MR483000SG3-EF, MPS484000HL (X can be A to Z, a to z, 0-9,
"-" or blank, indicating the customer code, which do not affect
safety and EMC)
Brand: **MEGMEET**
Sample Received Date: April, 07, 2024
Sample Status: Normal
Sample ID: SI2401097 A-NO.01
Date of Tested: From 2024-04-08 to 2024-05-06

APPLICABLE STANDARDS	
STANDARD	TEST RESULTS
ETSI EN 300 386 V2.2.1:2022 EN 55032:2015/A11:2020	Pass
EN IEC 61000-3-11:2019	Pass
EN 61000-3-12:2011	Pass
ETSI EN 300 386 V2.2.1:2022 EN 55035:2017/A11:2020	Pass

Prepared By:

A handwritten signature in black ink, appearing to read 'Wite Chen', positioned above a horizontal line.

Wite Chen

Engineer Project Associate

Checked By:

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Denny Huang

Senior Project Engineer

Approved By:

A handwritten signature in black ink, appearing to read 'Stephen Guo', positioned above a horizontal line.

Stephen Guo

Operations Manager

2. TEST METHODOLOGY

All tests were performed in accordance with the standard ETSI EN 300 386 V2.2.1:2022
EN 55032:2015/A11:2020, EN 55035:2017/A11:2020, EN IEC 61000-3-11:2019, EN 61000-3-12:2011.

3. FACILITIES AND ACCREDITATION

Accreditation Certificate	A2LA (Certificate No.: 4102.01) UL Verification Services (Guangzhou) Co., Ltd. Song Shan Lake Branch. has been assessed and proved to be in compliance with A2LA. FCC (FCC Designation No.: CN1187) UL Verification Services (Guangzhou) Co., Ltd. Song Shan Lake Branch. Has been recognized to perform compliance testing on equipment subject to the Commission's Declaration of Conformity (DoC) and Certification rules. ISED (Company No.: 21320) UL Verification Services (Guangzhou) Co., Ltd. Song Shan Lake Branch. has been registered and fully described in a report filed with ISED. The Company Number is 21320 and the test lab Conformity Assessment Body Identifier (CABID) is CN0046. VCCI (Registration No.: G-20192, C-20153, T-20155 and R-20202) UL Verification Services (Guangzhou) Co., Ltd. Song Shan Lake Branch. has been assessed and proved to be in compliance with VCCI, the Membership No. is 3793. Facility Name: Chamber D, the VCCI registration No. is G-20192 and R-20202 Shielding Room B, the VCCI registration No. is C-20153 and T-20155
Test lab Accreditation Certificate	A2LA (Certificate No.: 4893.01) DongGuan ShuoXin Electronic Technology Co., Ltd. has been assessed and proved to be in compliance with A2LA. CNAS (Certificate No.: CNAS L3098) DongGuan ShuoXin Electronic Technology Co., Ltd. has been assessed and proved to be in compliance with CNAS. CNAS (Certificate No.: CNAS L0579) Shenzhen Academy of Metrology and Quality Inspection National Digital Electronic Product testing Center. has been assessed and proved to be in compliance with CNAS. CNAS (Certificate No.: CNAS L0468) Guangdong Dongguan Quality Supervision Testing Center. has been assessed and proved to be in compliance with CNAS. CNAS (Certificate No.: CNAS L5053) Cerpess Technology (Dong Guan) Co., Ltd. has been assessed and proved to be in compliance with CNAS.
Test location	Conducted emissions test site: Guangdong Dongguan Quality Supervision Testing Center NO.2 South Industry Road, Dongguan Songshan Lake Sci.& Tech. Industrial Park, Guangdong Province. Voltage fluctuations and flicker and harmonic current test site: Shenzhen Academy of Metrology and Quality Inspection National Digital Electronic Product testing Center. NETC Building, No.4 tongfu Rd., Xili, Nanshan, Shenzhen China Continuous RF electromagnetic field disturbances test site: Cerpess Technology (Dong Guan) Co., Ltd. Room 102, No. 5, Xing'an Road, Chang'an Town, Dongguan City,

Guangdong 523850, China.

Other items are tested by the following companies:

DongGuan ShuoXin Electronic Technology Co., Ltd.

Zone A, 1F, No. 6, XinGang Road YuanGang Street, XinAn District, ChangAn Town, DongGuan City, GuangDong, China.

Remark:

- 1) The device described above was tested by DongGuan ShuoXin Electronic Technology Co., Ltd. to determine the maximum emission levels emanated from the device and severity levels of the device endure and its performance criterion. The measurement results are contained in this test report and DongGuan ShuoXin Electronic Technology Co., Ltd. assumes full responsibility for the accuracy and completeness of these measurements.
- 2) This report applies to the above sample only and shall not be reproduced in part without written approval of UL Verification Services (Guangzhou) Co., Ltd, Song Shan Lake Branch.

4. CALIBRATION AND UNCERTAINTY

4.1. MEASURING INSTRUMENT CALIBRATION

The measuring equipment utilized to perform the tests documented in this report has been calibrated in accordance with the manufacturer's recommendations and is traceable to recognized national standards.

4.2. MEASUREMENT UNCERTAINTY

Where relevant, the following measurement uncertainty levels have been estimated for tests performed on the apparatus:

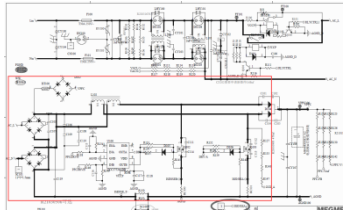
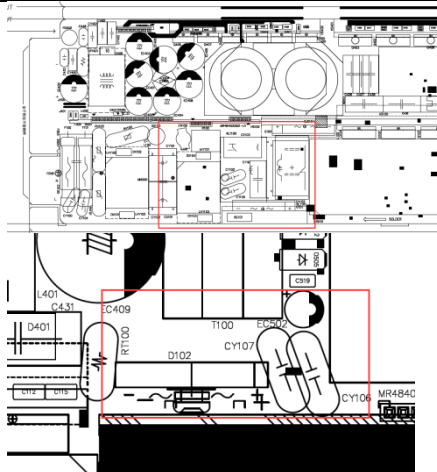
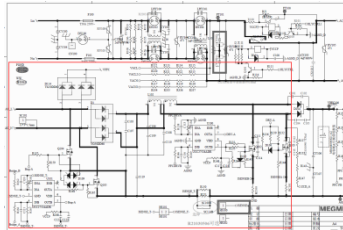
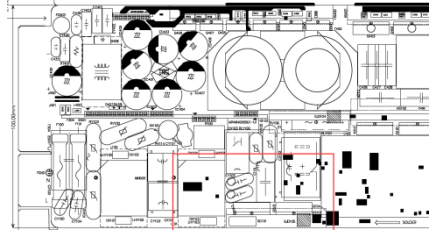
Test Item	Measurement Frequency Range	K	U(dB)
Conducted emissions (AC mains power ports)	0.15MHz - 30MHz	2	3.3
Radiated emissions below 1GHz	30MHz ~ 200MHz (H)	2	4.6
	30MHz ~ 200MHz (V)	2	4.6
	200MHz ~ 1000MHz(H)	2	5.08
	200MHz ~ 1000MHz(V)	2	6.10
Radiated emissions Above 1GHz	1000MHz ~ 6000MHz (H)	2	5.12
	1000MHz ~ 6000MHz (V)	2	5.12
Note1: This uncertainty represents an expanded uncertainty expressed at approximately the 95 % confidence level using a coverage factor of k=2.			

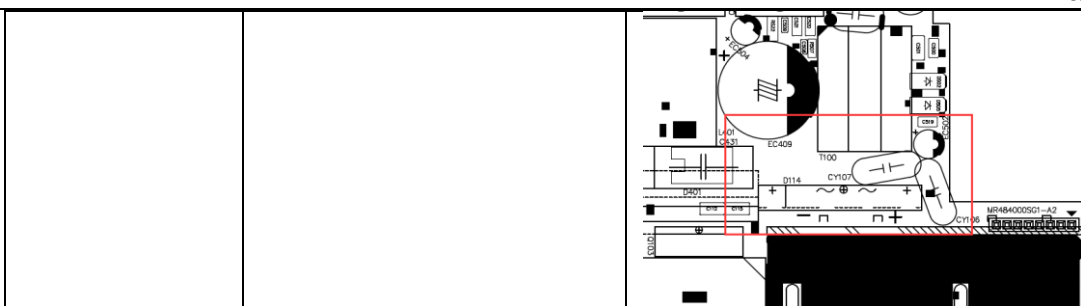
5. EQUIPMENT UNDER TEST

5.1. DESCRIPTION OF EUT

EUT Name	SWITCHING POWER SUPPLY
Model	MR484000SG2-GS
Series Model	MR484000SG1XXXXXX, MR484000SG2XXXXXX, MR483000SG3-EF, MPS484000HL (X can be A to Z, a to z, 0-9, "-" or blank, indicating the customer code, which do not affect safety and EMC)
Model Difference	The models are identical to each other except for model name, Circuit, PCB layout, input current rating and output rating. MR484000SG1XXXXXX and MPS484000HL are identical to each other except for model name. MR484000SG2XXXXXX and MR483000SG3-EF are identical to each other except for model name, input current rating and output rating. MR484000SG1XXXXXX and MR484000SG2XXXXXX are identical to each other except for model name, Circuit and PCB layout, see below Tabel A for details:
EUT Classification	Class I
Highest Internal Frequency	Below 108MHz
Ratings	Input: See below model list Table B for details. Output: See below model list Table B for details.

Tabel A:

Model	Circuit	PCB layout
MR484000SG1XXXXXX, MPS484000HL		
MR484000SG2XXXXXX, MR483000SG3-EF		



Tabel B:

Model	Input	Output (ambient 45°C)	Output (ambient 75°C)
MR484000SG1XXXXXX, MPS484000HL, MR484000SG2XXXXXX	100-130Vac, 50/60Hz, 21A	53.5Vdc, 32.5A	53.5Vdc, 14.9A
	200-250Vac, 50/60Hz, 21A	53.5Vdc, 75A	53.5Vdc, 37.38A
MR483000SG3-EF	100-130Vac, 50/60Hz, 17A	53.5Vdc, 25.2A	--
	200-250Vac, 50/60Hz, 17A	53.5Vdc, 56.1A	--

5.2. TEST MODE

Test Mode	Description
M01	FULL LOAD for model MR484000SG2-GS
M02	HALF LOAD
M03	NULL LOAD

5.3. MONITORING OF EUT FOR ALL IMMUNITY TEST

Other	Monitor output voltage via multimeters
-------	--

5.4. EUT ACCESSORY

Dummy load	
Model No.:	/
Input:	/
Output:	/
DC Cable:	1.0 Meter without ferrite, unshielded

5.5. SUPPORT UNITS FOR SYSTEM TEST

The following support units or accessories were used to form a representative test configuration during the tests.

Item	Accessory	Brand Name	Model Name	Specification	Description
1	Dummy Load	N/A	N/A	2Ω, 2000W	ATT support
2	Dummy Load	N/A	N/A	4Ω, 1200W	ATT support
3	Dummy Load	N/A	N/A	4Ω, 1200W	ATT support
4	Dummy Load	N/A	N/A	4Ω, 1200W	ATT support
5	Dummy Load	N/A	N/A	5Ω, 1200W	ATT support

The following cables were used to form a representative test configuration during the tests.

Item	Type of cable	Shielded Type	Ferrite Core	Specification
C-1	DC Cable	NO	NO	0.5m
C-2	AC Cable	NO	NO	1.5m

6. MEASURING EQUIPMENT AND SOFTWARE USED

Test Equipment of Conducted emissions (AC mains power ports)					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
EMI Test Receiver	R&S	ESR3	DQM1179	April. 26, 2024	April. 25, 2025
LISN	R&S	ENV4200	DQM1182	April. 26, 2024	April. 25, 2025
Measurement Software	Farad	EZ-EMC (Ver.ATT-03A)	N/A	N/A	N/A

Test Equipment of Radiated emissions below 1GHz					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
Log-Bicon Antenna	SCHWARZBEC K	VULB9168	VULB9168-192	Mar. 30, 2024	Mar. 29, 2025
EMI Test Receiver	R&S	ESCI	101307	Nov. 30, 2023	Nov. 29, 2024
Pre-Amplifier	HP	8447F	3113A05680	Mar. 30, 2024	Mar. 29, 2025
Turn Table	UC	UC3000	N/A	N/A	N/A
Antenna Mast	UC	UC3000	N/A	N/A	N/A
Measurement Software	Farad	EZ-EMC (Ver.ATT-03A)	N/A	N/A	N/A

Test Equipment of Voltage fluctuations and flicker					
Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Calibrated until
1	Harmonic scintillation analyzer and impedance network	CI	CTSHL SYSTEM FOR MX90	SB7808	2024-10-31
2	power source	CI	MX90-3Pi-MB-400-411-413-LF	SB9555/04	2024-10-31

Test Equipment of Electrostatic Discharge					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
Electrostatic discharge generator	Prima	ESD61002TC	PRI18032186 7	May. 24, 2023	May. 23, 2024

Test Equipment of Continuous RF electromagnetic field disturbances					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
Signal Generator	R&S	SML03	103287	Aug. 03, 2023	Aug. 02, 2024
Signal Generator	R&S	SMR30	100049	Aug. 03, 2023	Aug. 02, 2024
Power Sensor	R&S	NRP-Z11	106374	Jun. 13, 2023	Jun. 12, 2024
Power Amplifier	BONN	BLWA0830-160/100/40D	76659	Aug. 03, 2023	Aug. 02, 2024
Preamplifier	MILMEGA	AS1860-30	10040456	Aug. 03, 2023	Aug. 02, 2024
Istropic Electric Field Probe	EST.LINDGREN	HI-6105	137445	Aug. 03, 2023	Aug. 02, 2024
EMS Antenna	R&S	HL046E	100028	N/A	N/A
Broad-Band Horn Antenna	Schwarzbeck	BBHA9120 E	475	Jan. 07, 2024	Jan. 06, 2025
Temperature/Humidity Meter	mingle	ETH529	N/A	Jan. 03, 2024	Jan. 02, 2025

Test Equipment of Electrical fast transients burst (AC mains power ports)					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
Transient 3000 Test System	EMC-Partner	TRA3000	F-D-V-1501	Nov. 30, 2023	Nov. 29, 2024
Capacitive Coupling Clamp	EMC-Partner	CN-EFT1000	709	May. 16, 2023	May. 15, 2024

Test Equipment of Surges (AC mains power ports)					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
Lightning Surge Generator	Prima	SUG61005BG	PR13015543	Nov. 07, 2023	Nov. 06, 2024

Test Equipment of Continuous induced RF disturbances (AC mains power ports)					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
Conducted Immunity Test System	Frankonia	CIT-10	126B1210	Nov. 30, 2023	Nov. 29, 2024
CDN	CD (Compliance Direction Systems Inc)	CDN M2+M3	3588	Nov. 30, 2023	Nov. 29, 2024
Attenuation	TESEQ	ATN 6075(6dB 75W)	32126	Nov. 07, 2023	Nov. 06, 2024

Test Equipment of Voltage dips and interruptions (AC mains power ports)					
Equipment	Manufacturer	Model No.	Serial No.	Last Cal.	Due Date
Transient 3000 Test System	EMC-Partner	TRA3000	F-D-V-1501	Nov. 30, 2023	Nov. 29, 2024

7. EMISSION TEST

7.1. CONDUCTED EMISSIONS (AC MAINS POWER PORTS)

LIMITS

(a.) Limits of conducted emissions from the AC mains power ports of Class A equipment

Frequency range MHz	Coupling device	Detector type / bandwidth	Limit dBuV
0.15 to 0.5	AMN	Quasi Peak / 9 kHz	79
0.5 to 30			73
0.15 to 0.5	AMN	Average / 9 kHz	66
0.5 to 30			60

(b.) Limits of conducted emissions from the AC mains power ports of Class B equipment

Frequency range MHz	Coupling device	Detector type / bandwidth	Limit dBuV
0.15 to 0.5	AMN	Quasi Peak / 9 kHz	66 to 56
0.5 to 5			56
5 to 30			60
0.15 to 0.5	AMN	Average / 9 kHz	56 to 46
0.5 to 5			46
5 to 30			50

(c.) Limits of asymmetric mode conducted emissions of Class A equipment

Frequency range MHz	Coupling device	Detector type / bandwidth	Limit dBuV	Limit dBuA
0.15 -0.5	AAN	Quasi Peak / 9 kHz	97 to 87	n/a
0.5 -30			87	n/a
0.15 -0.5	AAN	Average / 9 kHz	84 to 74	n/a
0.5 -30			74	n/a
0.15 -0.5	Current Probe	Quasi Peak / 9 kHz	n/a	53 to 43
0.5 -30			n/a	43
0.15 -0.5	Current Probe	Average / 9 kHz	n/a	40 to 30
0.5 -30			n/a	30

(d.) Limits of asymmetric mode conducted emissions of Class B equipment

Frequency range MHz	Coupling device	Detector type / bandwidth	Limit dBuV	Limit dBuA
0.15 -0.5	AAN	Quasi Peak / 9 kHz	84 to 74	n/a
0.5 -30			74	n/a
0.15 -0.5	AAN	Average / 9 kHz	74 to 64	n/a
0.5 -30			64	n/a
0.15 -0.5	Current Probe	Quasi Peak / 9 kHz	n/a	40 to 30
0.5 -30			n/a	30
0.15 -0.5	Current Probe	Average / 9 kHz	n/a	30 to 20
0.5 -30			n/a	20

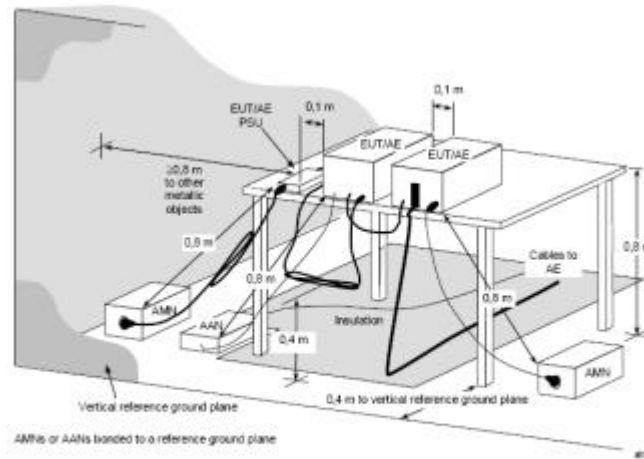
Note:

- 1) The tighter limit applies at the band edges.
- 2) The limit of " * " marked band means the limitation decreases linearly with the logarithm of the frequency in the range.

TEST PROCEDURE

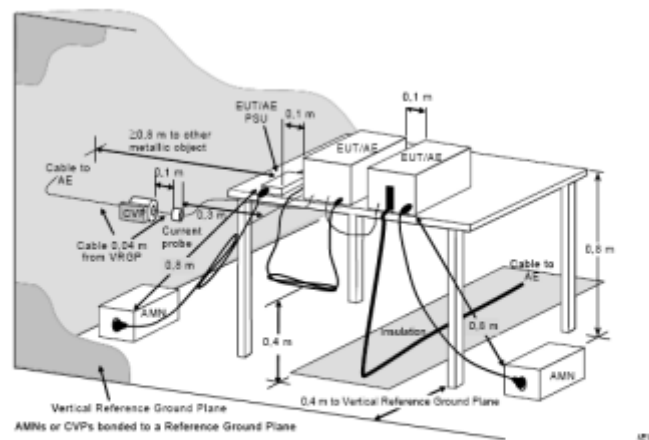
- 1) The EUT was placed on the top of a rotating table 0.8 meters above the horizontal ground plane and being connected to the power mains through a line impedance stabilization network (LISN). All other support equipment powered from additional LISN(s). The LISN provide 50 ohm/ 50uH of coupling impedance for the measuring instrument.
- 2) Interconnecting cables that hang closer than 40 cm to the ground plane shall be folded back and forth in the center forming a bundle 30 to 40 cm long.
- 3) I/O cables that are not connected to a peripheral shall be bundled in the center. The end of the cable may be terminated, if required, using the correct terminating impedance. The overall length shall not exceed 1 m.
- 4) Cables of hand-operated devices, such as keyboards and mice, shall be placed as for normal used.
- 5) LISN at least 80 cm from nearest part of EUT chassis.
- 6) Conducted emissions from the EUT measured in the frequency range between 0.15 MHz and 30MHz using CISPR Quasi-Peak and average detector mode.

TEST SETUP



The 0,5 m distance specified between EUT/AE/PSU and AMN/AAN, is applicable only to the EUT being measured. If the device is AE then it shall be $\geq 0,5$ m.

a) Example measurement arrangement for table-top EUT (alternative 1)



The 0,8 m distance specified between EUT/local AE/PSU and AMN/AAN, is applicable only to the EUT being measured. If the device is AE then it shall be $\geq 0,8$ m.

b) Example measurement arrangement for table-top EUT measuring in accordance with C.4.1.6.4

TEST ENVIRONMENT

Temperature	26.1°C	Relative Humidity	54.1%
Atmosphere Pressure	101kPa		

TEST DATE / ENGINEER

Test Date	April. 26, 2024	Test By	Vian
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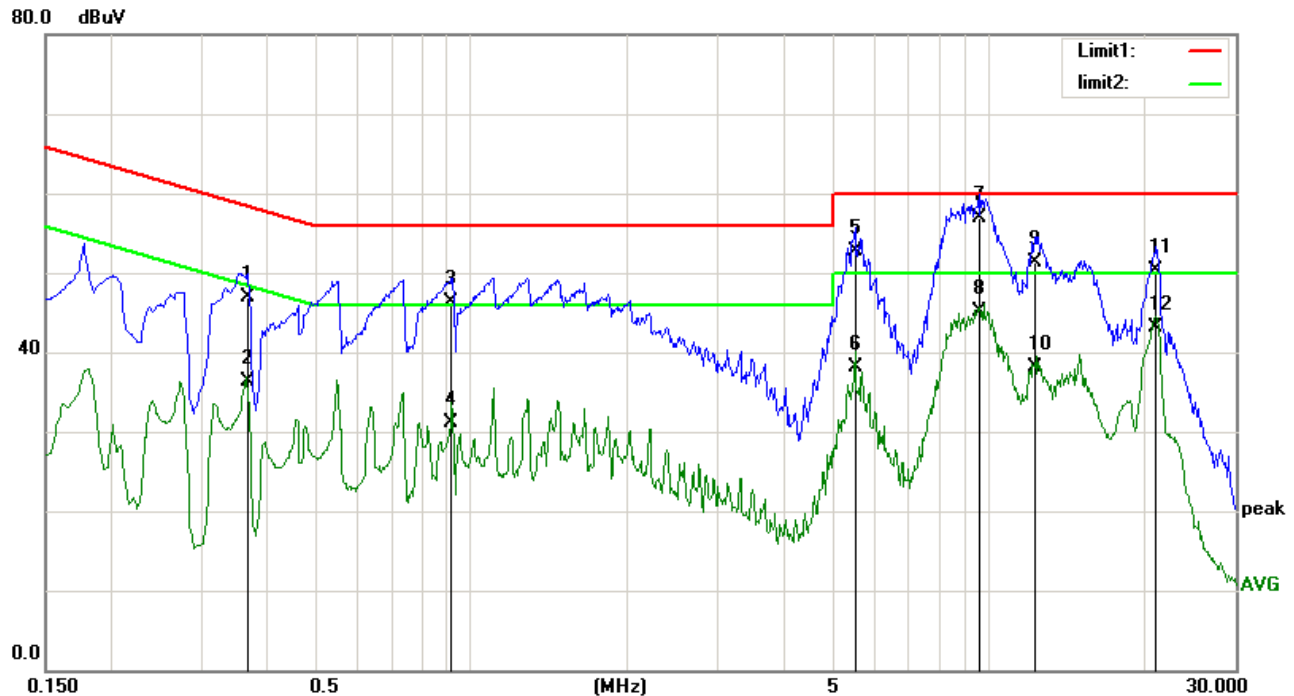
TEST MODE

Pre-test Mode:	M01 ~ M03
Final Test Mode:	M01

Note: All test modes had been tested, but only the worst data recorded in the report.

TEST RESULTS

Test Mode:	M01	Line:	Line
Test Model:	MR484000SG2-GS	Test Voltage:	AC 230V/50Hz
Note:	53.5V/75A		

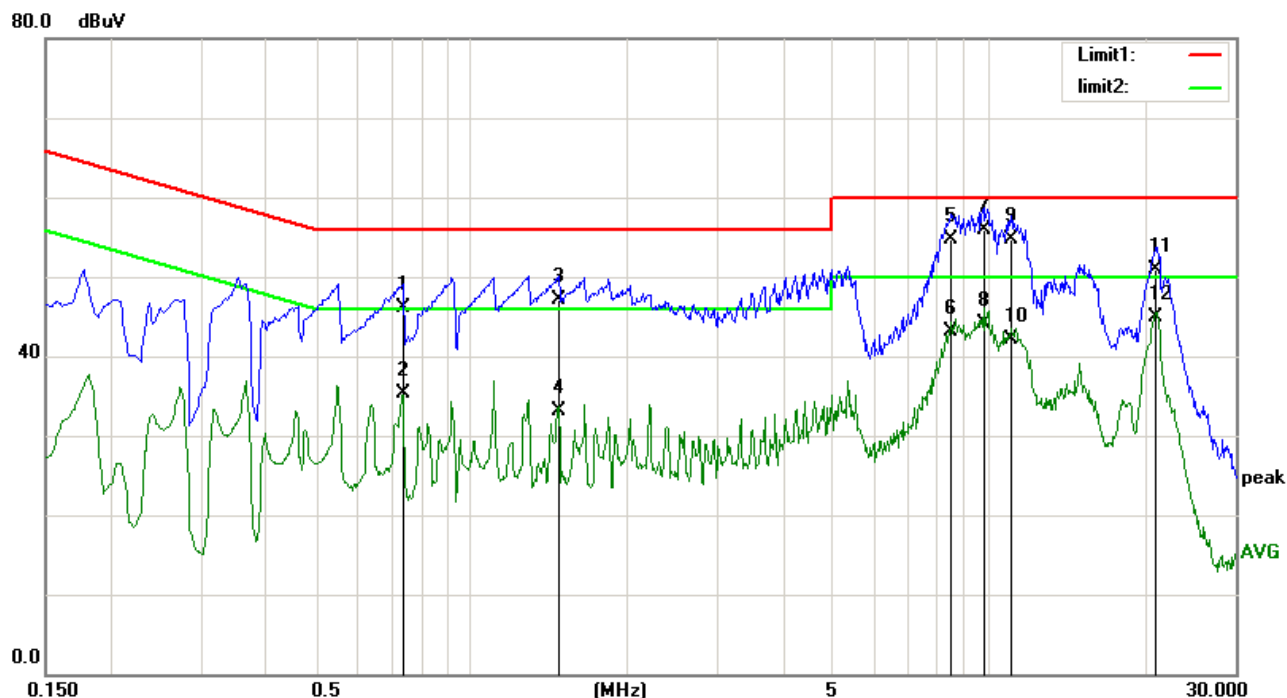


No.	Frequency (MHz)	Reading (dBuV)	Correct Factor(dB)	Result (dBuV)	Limit (dBuV)	Margin (dB)	Remark
1	0.3700	36.11	10.87	46.98	58.50	-11.52	QP
2	0.3700	25.42	10.87	36.29	48.50	-12.21	AVG
3	0.9180	35.55	10.77	46.32	56.00	-9.68	QP
4	0.9180	20.29	10.77	31.06	46.00	-14.94	AVG
5	5.5219	41.86	10.82	52.68	60.00	-7.32	QP
6	5.5219	27.34	10.82	38.16	50.00	-11.84	AVG
7	9.6019	46.00	10.92	56.92	60.00	-3.08	QP
8	9.6019	34.24	10.92	45.16	50.00	-4.84	AVG
9	12.2939	40.31	10.95	51.26	60.00	-8.74	QP
10	12.2939	27.21	10.95	38.16	50.00	-11.84	AVG
11	21.1020	39.32	11.04	50.36	60.00	-9.64	QP
12	21.1020	32.14	11.04	43.18	50.00	-6.82	AVG

Remark: Result = Reading +Correct (Insertion Loss + Cable Loss + Attenuator Factor)

Margin = Result - Limit

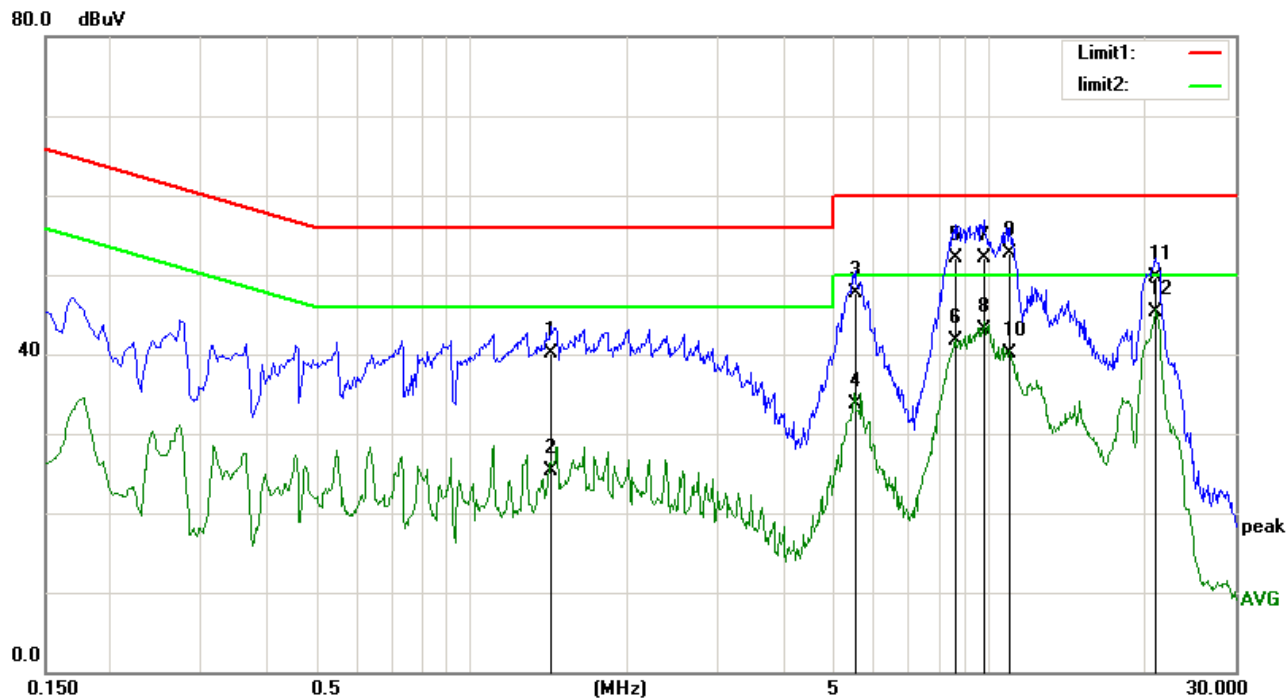
Test Mode:	M01	Line:	Neutral
Test Model:	MR484000SG2-GS	Test Voltage:	AC 230V/50Hz
Note:	53.5V/75A		



No.	Frequency (MHz)	Reading (dBuV)	Correct Factor(dB)	Result (dBuV)	Limit (dBuV)	Margin (dB)	Remark
1	0.7380	35.36	10.79	46.15	56.00	-9.85	QP
2	0.7380	24.47	10.79	35.26	46.00	-10.74	AVG
3	1.4780	36.31	10.75	47.06	56.00	-8.94	QP
4	1.4780	22.34	10.75	33.09	46.00	-12.91	AVG
5	8.4459	43.84	10.94	54.78	60.00	-5.22	QP
6	8.4459	32.22	10.94	43.16	50.00	-6.84	AVG
7	9.7858	44.95	10.92	55.87	60.00	-4.13	QP
8	9.7858	33.24	10.92	44.16	50.00	-5.84	AVG
9	11.0778	43.83	10.93	54.76	60.00	-5.24	QP
10	11.0778	31.16	10.93	42.09	50.00	-7.91	AVG
11	21.0740	39.82	11.04	50.86	60.00	-9.14	QP
12	21.0740	33.94	11.04	44.98	50.00	-5.02	AVG

Remark: Result = Reading +Correct (Insertion Loss + Cable Loss + Attenuator Factor)
Margin = Result – Limit

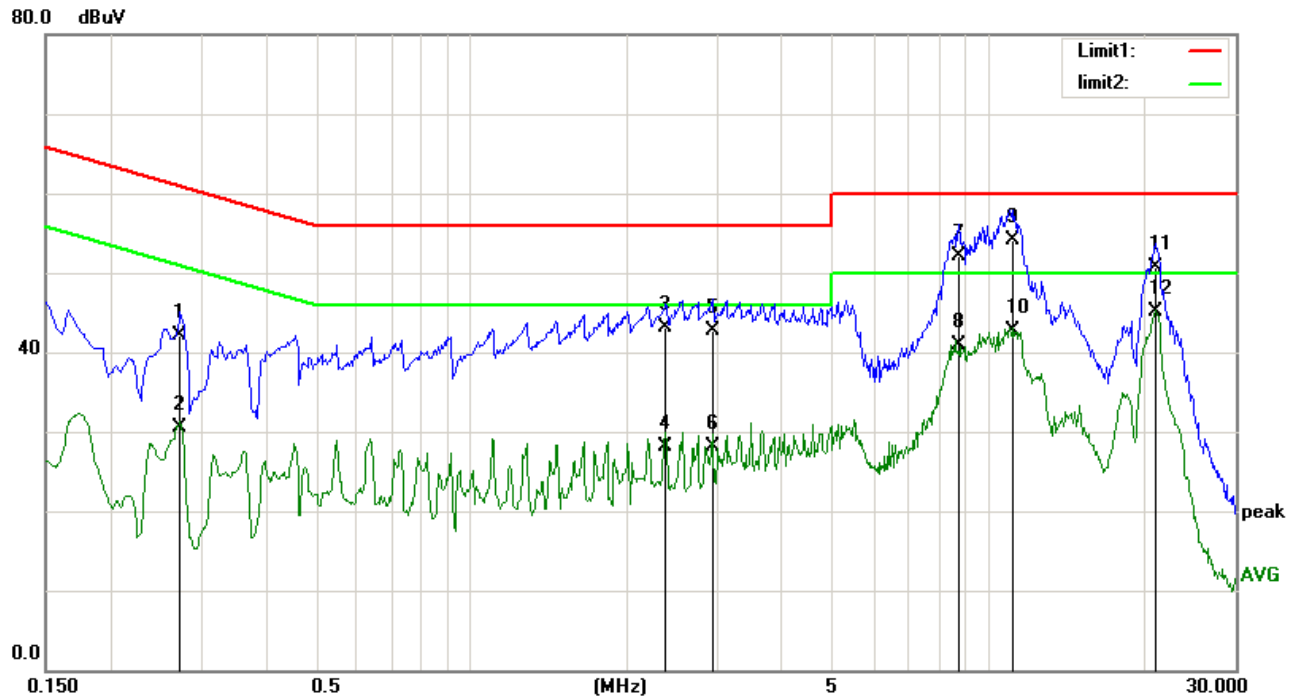
Test Mode:	M01	Line:	Line
Test Model:	MR484000SG2-GS	Test Voltage:	AC 110V/60Hz
Note:	53.5V/32.5A		



No.	Frequency (MHz)	Reading (dBuV)	Correct Factor(dB)	Result (dBuV)	Limit (dBuV)	Margin (dB)	Remark
1	1.4299	29.42	10.74	40.16	56.00	-15.84	QP
2	1.4299	14.60	10.74	25.34	46.00	-20.66	AVG
3	5.5059	36.83	10.82	47.65	60.00	-12.35	QP
4	5.5059	22.86	10.82	33.68	50.00	-16.32	AVG
5	8.6219	41.21	10.95	52.16	60.00	-7.84	QP
6	8.6219	30.81	10.95	41.76	50.00	-8.24	AVG
7	9.7899	41.24	10.92	52.16	60.00	-7.84	QP
8	9.7899	32.26	10.92	43.18	50.00	-6.82	AVG
9	11.0099	41.74	10.93	52.67	60.00	-7.33	QP
10	11.0099	29.23	10.93	40.16	50.00	-9.84	AVG
11	20.9900	38.64	11.04	49.68	60.00	-10.32	QP
12	20.9900	34.22	11.04	45.26	50.00	-4.74	AVG

Remark: Result = Reading +Correct (Insertion Loss + Cable Loss + Attenuator Factor)
Margin = Result – Limit

Test Mode:	M01	Line:	Neutral
Test Model:	MR484000SG2-GS	Test Voltage:	AC 110V/60Hz
Note:	53.5V/32.5A		



No.	Frequency (MHz)	Reading (dBuV)	Correct Factor(dB)	Result (dBuV)	Limit (dBuV)	Margin (dB)	Remark
1	0.2740	31.57	10.59	42.16	60.99	-18.83	QP
2	0.2740	19.97	10.59	30.56	50.99	-20.43	AVG
3	2.3820	32.31	10.81	43.12	56.00	-12.88	QP
4	2.3820	17.35	10.81	28.16	46.00	-17.84	AVG
5	2.9260	31.94	10.74	42.68	56.00	-13.32	QP
6	2.9260	17.42	10.74	28.16	46.00	-17.84	AVG
7	8.7539	41.25	10.94	52.19	60.00	-7.81	QP
8	8.7539	30.01	10.94	40.95	50.00	-9.05	AVG
9	11.1618	43.23	10.93	54.16	60.00	-5.84	QP
10	11.1618	31.71	10.93	42.64	50.00	-7.36	AVG
11	21.0220	39.75	11.04	50.79	60.00	-9.21	QP
12	21.0220	34.02	11.04	45.06	50.00	-4.94	AVG

Remark: Result = Reading +Correct (Insertion Loss + Cable Loss + Attenuator Factor)

Margin = Result – Limit

7.2. RADIATED EMISSIONS BELOW 1GHZ

LIMITS

Frequency (MHz)	Class A		Class B	
	At 10 m	At 3 m	At 10 m	At 3 m
	dB μ V/m	dB μ V/m	dB μ V/m	dB μ V/m
30 – 230	40	50	30	40
230 – 1000	47	57	37	47

Note:

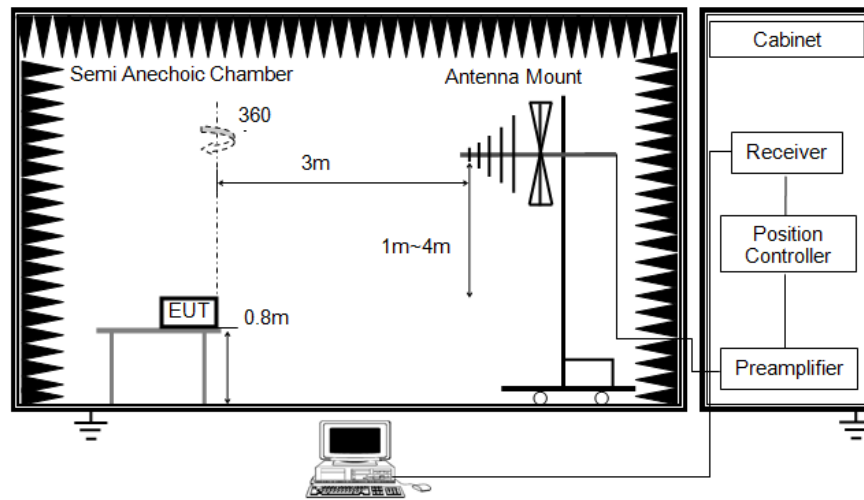
- 1) The limit for radiated test was performed according to CISPR 32.
- 2) The tighter limit applies at the band edges.
- 3) Emission level (dB μ V/m)=20log Emission level (μ V/m).
- 4) If the highest frequency of the internal sources of the EUT is less than 108 MHz, the measurement shall only be made up to 1 GHz. If the highest frequency of the internal sources of the EUT is between 108 MHz and 500 MHz, the measurement shall only be made up to 2 GHz.

If the highest frequency of the internal sources of the EUT is between 500 MHz and 1 GHz, the measurement shall only be made up to 5 GHz. If the highest frequency of the internal sources of the EUT is above 1 GHz, the measurement shall be made up to 5 times the highest frequency or 6 GHz, whichever is less.

TEST PROCEDURE

- 1) The EUT was arranged to its worst case and then tune the antenna tower (from 1 m to 4 m) and turntable (from 0 degree to 360 degrees) to find the maximum reading. A pre-amp was used for the test in order to get better signal level. Both horizontal and vertical polarizations of the antenna are set to make the measurement.
- 2) The EUT was placed on a turntable with 80 cm above ground.
- 3) Interconnecting cables that hang closer than 40 cm to the ground plane shall be folded back and forth in the center forming a bundle 30 to 40 cm long.
- 4) I/O cables that are not connected to a peripheral shall be bundled in the center. The end of the cable may be terminated, if required, using the correct terminating impedance. The overall length shall not exceed 1 m.
- 5) Cables of hand-operated devices, such as keyboards and mice, shall be placed as for normal used.
- 6) The EUT was set 3 meters from the interference receiving antenna, which was mounted on the top of a variable height antenna tower.
- 7) For measurement below 1 GHz, the initial step in collecting Radiated emission data is a spectrum analyzer peak detector mode pre-scanning the measurement frequency range. Significant peaks are then marked and then Quasi Peak detector mode re-measured.

TEST SETUP



Below 1 GHz and above 30 MHz

TEST ENVIRONMENT

Temperature	24.8°C	Relative Humidity	53%
Atmosphere Pressure	1.1kPa		

TEST DATE / ENGINEER

Test Date	April. 24, 2024	Test By	vian
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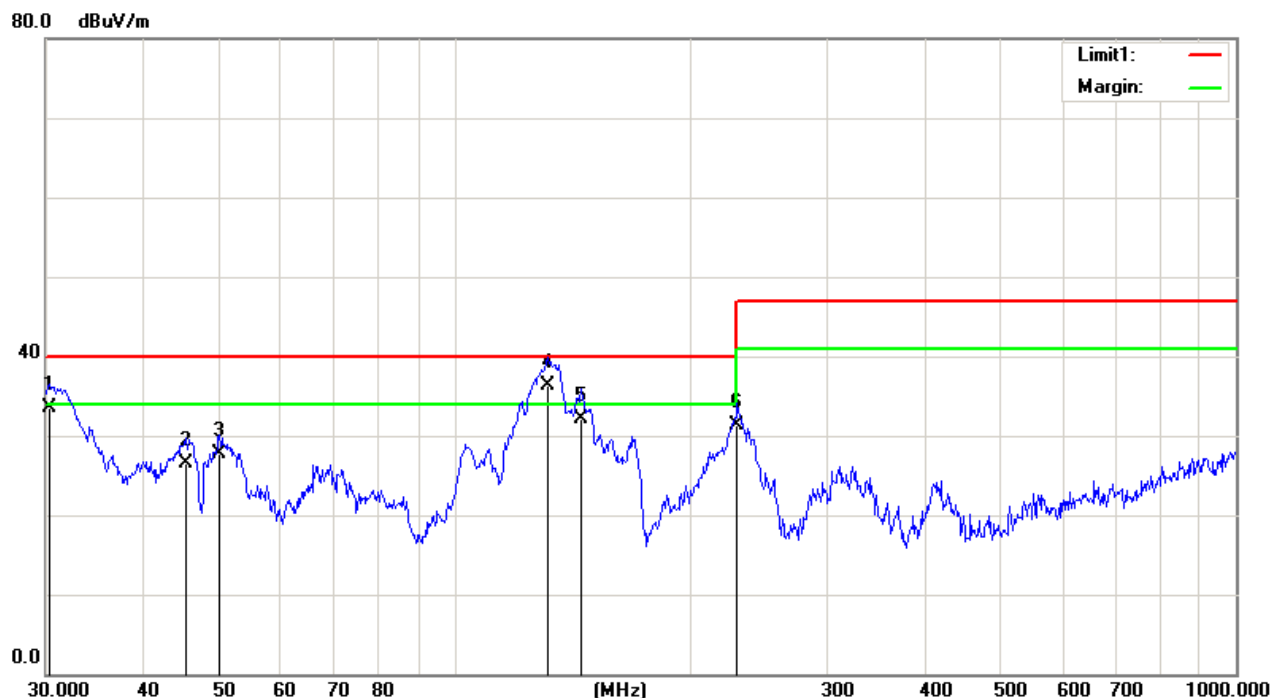
TEST MODE

Pre-test Mode:	M01 ~ M03
Final Test Mode:	M01

Note: All test modes had been tested, but only the worst data recorded in the report.

TEST RESULTS

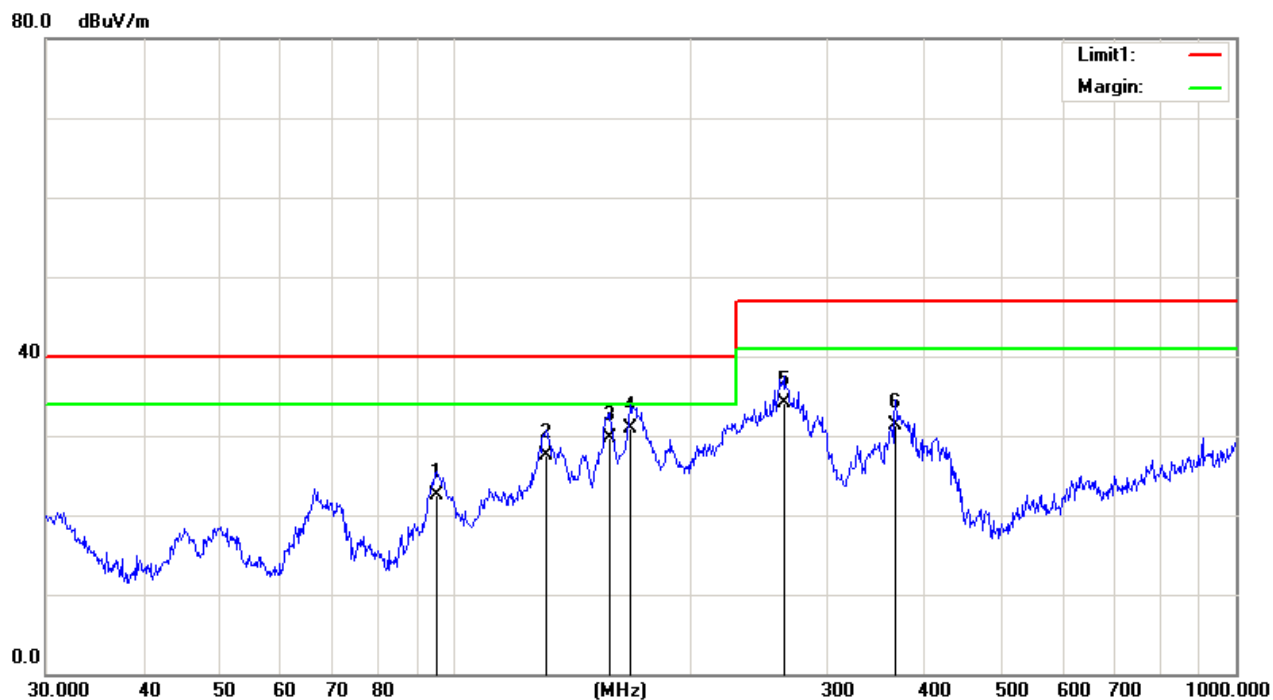
Test Mode:	M01	Polarity:	Vertical
Test Model:	MR484000SG2-GS	Test Voltage:	AC230V_50Hz
Note:	53.5V/75A		



No.	Frequency (MHz)	Reading (dBuV/m)	Correct Factor(dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Remark
1	30.3172	46.57	-12.98	33.59	40.00	-6.41	QP
2	45.3755	40.91	-14.46	26.45	40.00	-13.55	QP
3	50.0566	39.99	-12.34	27.65	40.00	-12.35	QP
4	131.7576	50.20	-13.95	36.25	40.00	-3.75	QP
5	145.3505	45.54	-13.39	32.15	40.00	-7.85	QP
6	230.0985	40.40	-9.11	31.29	47.00	-15.71	QP

Note: 1. Result = Reading +Correct (Amplifier Factor + Cable Loss + Antenna Factor)
2. Margin = Result - Limit

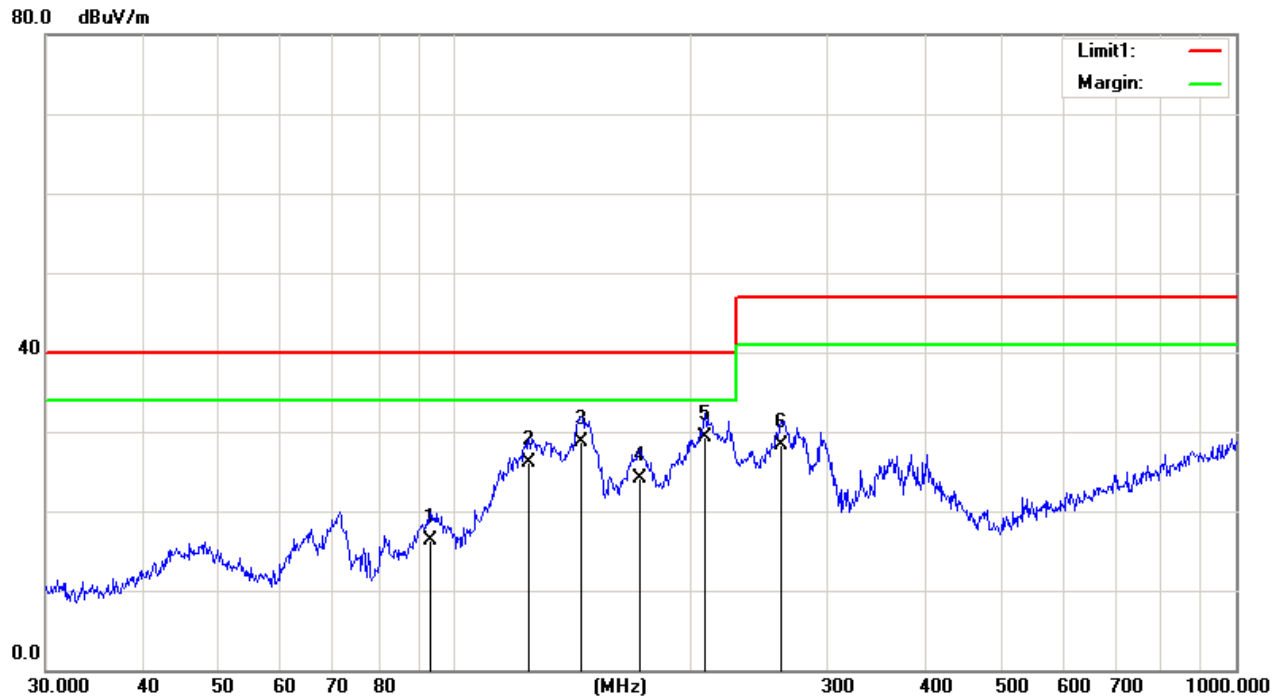
Test Mode:	M01	Polarity:	Horizontal
Test Model:	MR484000SG2-GS	Test Voltage:	AC230V_50Hz
Note:	53.5V/75A		



No.	Frequency (MHz)	Reading (dBuV/m)	Correct Factor(dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Remark
1	94.7600	40.62	-18.03	22.59	40.00	-17.41	QP
2	131.2965	42.47	-15.01	27.46	40.00	-12.54	QP
3	158.1123	43.78	-14.13	29.65	40.00	-10.35	QP
4	167.8242	43.57	-12.76	30.81	40.00	-9.19	QP
5	263.8190	40.09	-5.91	34.18	47.00	-12.82	QP
6	366.8231	40.73	-9.48	31.25	47.00	-15.75	QP

Note: 1. Result = Reading +Correct (Amplifier Factor + Cable Loss + Antenna Factor)
2. Margin = Result – Limit

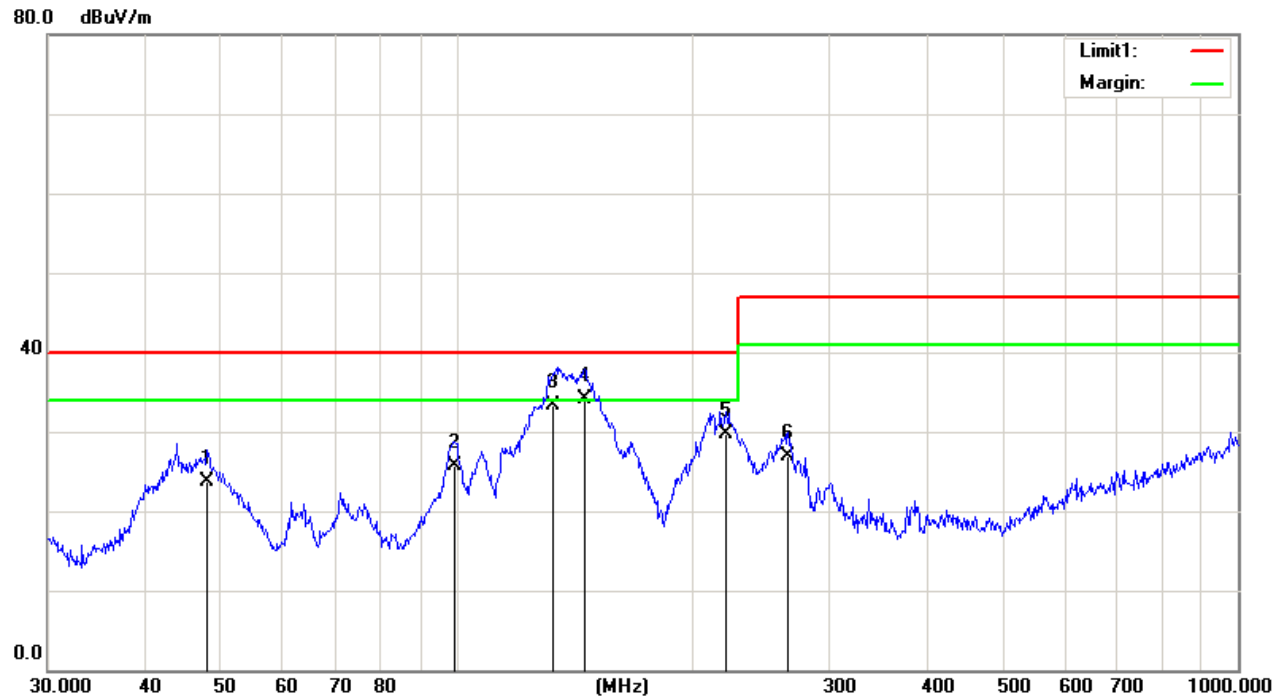
Test Mode:	M01	Polarity:	Horizontal
Test Model:	MR484000SG2-GS	Test Voltage:	AC110V_60Hz
Note:	53.5V/32.5A		



No.	Frequency (MHz)	Reading (dBuV/m)	Correct Factor(dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Remark
1	93.1132	34.58	-18.23	16.35	40.00	-23.65	QP
2	124.5690	40.40	-14.35	26.05	40.00	-13.95	QP
3	145.3505	42.97	-14.32	28.65	40.00	-11.35	QP
4	172.5988	36.22	-12.04	24.18	40.00	-15.82	QP
5	209.3129	39.63	-10.37	29.26	40.00	-10.74	QP
6	261.9753	34.23	-5.99	28.24	47.00	-18.76	QP

Note: 1. Result = Reading +Correct (Amplifier Factor + Cable Loss + Antenna Factor)
2. Margin = Result – Limit

Test Mode:	M01	Polarity:	Vertical
Test Model:	MR484000SG2-GS	Test Voltage:	AC110V_60Hz
Note:	53.5V/32.5A		



No.	Frequency (MHz)	Reading (dBuV/m)	Correct Factor(dB/m)	Result (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Remark
1	47.9940	37.28	-13.54	23.74	40.00	-16.26	QP
2	99.5279	41.33	-15.59	25.74	40.00	-14.26	QP
3	132.3792	47.13	-13.85	33.28	40.00	-6.72	QP
4	145.8608	47.51	-13.33	34.18	40.00	-5.82	QP
5	221.3921	40.85	-11.20	29.65	40.00	-10.35	QP
6	265.6757	36.74	-9.84	26.90	47.00	-20.10	QP

Note: 1. Result = Reading +Correct (Amplifier Factor + Cable Loss + Antenna Factor)
2. Margin = Result – Limit

7.3. HARMONIC CURRENT EMISSIONS LIMITS

EN61000-3-2 emission limits

Equipment Category	Harmonic Order	Max. Permissible Harmonic Current	Equipment Category	Harmonic Order	Max. Permissible Harmonic Current	
	n	A		n	A	mA/w
Class A	Odd Harmonics		Class D	Odd Harmonics only		
	3	2.30		3	2.30	3.4
	5	1.14		5	1.14	1.9
	7	0.77		7	0.77	1.0
	9	0.40		9	0.40	0.5
	11	0.33		11	0.33	0.35
	13	0.21		13	0.21	0.30
	15≤n≤39	0.15 x 15/n		15≤n≤39	0.15 x 15/n	3.85/n
	Even Harmonics					
	2	1.08				
	4	0.43				
	6	0.30				
	8≤n≤40	0.23 x 8/n				

IEC 61000-3-12 emission limits

Minimum R _{sce}	Admissible individual harmonic current I _h /I _{ref} ^a %						Admissible harmonic parameters %	
	I ₃	I ₅	I ₇	I ₉	I ₁₁	I ₁₃	THC/ I _{ref}	PWHC / I _{ref}
33	21.6	10.7	7.2	3.8	3.1	2	23	23
66	24	13	8	5	4	3	26	26
120	27	15	10	6	5	4	30	30
250	35	20	13	9	8	6	40	40
≥350	41	24	15	12	10	8	47	47

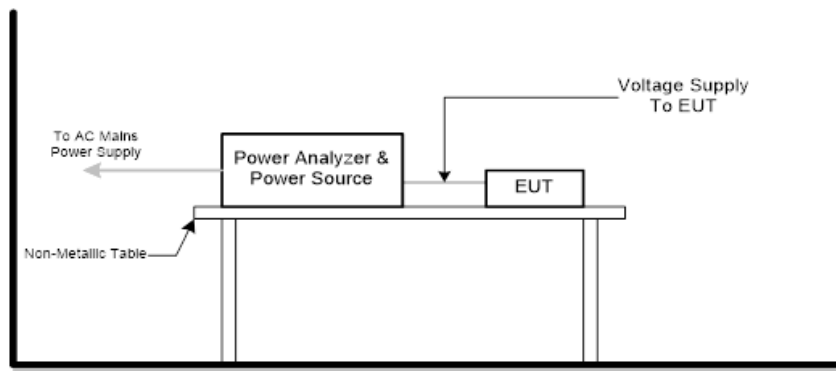
The relative values of even harmonics up to order 12 shall not exceed 16/h %. Even harmonics above order 12 are taken into account in THC and PWHC in the same way as odd order harmonics. Linear interpolation between successive R_{sce} values is permitted.

a_{Iref} = reference current; I_h = harmonic current component.

TEST PROCEDURE

- 1) The EUT was placed on the top of a wooden table 0.8 meters above the ground and operated to produce the maximum harmonic components under normal operating Condition.
- 2) The classification of EUT is according to EN 61000-3-2. The EUT is classified as follows:
Class A:
Balanced three-phase equipment, Household appliances excluding equipment as Class D, Tools excluding portable tools, Dimmers for incandescent lamps, audio equipment, equipment not specified in one of the three other classes.
Class B:
Portable tools. Arc welding equipment which is not professional equipment.
Class C:
Lighting equipment.
Class D:
Equipment having a specified power less than or equal to 600W of the following types: Personal computers and personal computer monitors and television receivers.
- 3) The correspondent test program of test instrument to measure the current harmonics emanated from EUT is chosen. The measure time shall be not less than the time necessary for the EUT to be exercised.

TEST SETUP



TEST ENVIRONMENT

Temperature	24.1°C	Relative Humidity	55.2%
Atmosphere Pressure	101kPa		

TEST DATE / ENGINEER

Test Date	Jan. 22, 2024	Test By	Vian
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TEST MODE

Pre-test Mode:	M01
Final Test Mode:	M01

Note: All test modes had been tested, but only the worst data recorded in the report.

TEST RESULTS

Test Mode:	M01	Test Voltage:	AC 230V_50Hz
Test Model:	MR484000SG2-GS		

Test Result: Pass → Measured I_{ref}: 17.259(Amps) → Source: Normal
 THC/I_{ref} (%): 0.0 → Limit (%): 22.0 → PWHC/I_{ref} (%): 0.0 → PWHC Limit (%): 22.0

Highest parameter values during test:

V_{RMS} (Volts): 230.71 → Frequency (Hz): 50.00
 I_{Peak} (Amps): 26.121 → I_{RMS} (Amps): 17.340
 I_{Fund} (Amps): 17.157(avg) → Crest Factor: 1.593
 Power (Watts): 3991 → Power Factor: 0.999

Harm #	Harm s(avg)	100%Limit	%of Limit	Harm s(max)	150%Limit	%of Limit	Status
→ 2	→ 0.017	→ 1.381	→ 1.4	→ 0.025	→ 2.071	→ 1.2	Pass
→ 3	→ 0.108	→ 3.728	→ 2.8	→ 0.118	→ 5.592	→ 2.1	Pass
→ 4	→ 0.002	→ 0.690	→ N/A	→ 0.004	→ 1.036	→ N/A	Pass
→ 5	→ 0.114	→ 1.847	→ 5.8	→ 0.120	→ 2.770	→ 4.2	Pass
→ 6	→ 0.006	→ 0.460	→ N/A	→ 0.003	→ 0.690	→ N/A	Pass
→ 7	→ 0.076	→ 1.243	→ 6.1	→ 0.076	→ 1.864	→ 4.1	Pass
→ 8	→ 0.005	→ 0.345	→ N/A	→ 0.004	→ 0.518	→ N/A	Pass
→ 9	→ 0.073	→ 0.656	→ 10.7	→ 0.071	→ 0.984	→ 7.4	Pass
→ 10	→ 0.002	→ 0.276	→ N/A	→ 0.003	→ 0.414	→ N/A	Pass
→ 11	→ 0.047	→ 0.535	→ 8.2	→ 0.049	→ 0.803	→ 5.1	Pass
→ 12	→ 0.003	→ 0.230	→ N/A	→ 0.004	→ 0.345	→ N/A	Pass
→ 13	→ 0.047	→ 0.345	→ 12.4	→ 0.040	→ 0.518	→ 8.7	Pass
→ 14	→ 0.003	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 15	→ 0.036	→ N/A	→ N/A	→ 0.036	→ N/A	→ N/A	N/A
→ 16	→ 0.003	→ N/A	→ N/A	→ 0.003	→ N/A	→ N/A	N/A
→ 17	→ 0.041	→ N/A	→ N/A	→ 0.049	→ N/A	→ N/A	N/A
→ 18	→ 0.003	→ N/A	→ N/A	→ 0.006	→ N/A	→ N/A	N/A
→ 19	→ 0.014	→ N/A	→ N/A	→ 0.018	→ N/A	→ N/A	N/A
→ 20	→ 0.001	→ N/A	→ N/A	→ 0.006	→ N/A	→ N/A	N/A
→ 21	→ 0.030	→ N/A	→ N/A	→ 0.034	→ N/A	→ N/A	N/A
→ 22	→ 0.001	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 23	→ 0.016	→ N/A	→ N/A	→ 0.019	→ N/A	→ N/A	N/A
→ 24	→ 0.003	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 25	→ 0.025	→ N/A	→ N/A	→ 0.029	→ N/A	→ N/A	N/A
→ 26	→ 0.002	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 27	→ 0.022	→ N/A	→ N/A	→ 0.028	→ N/A	→ N/A	N/A
→ 28	→ 0.002	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 29	→ 0.035	→ N/A	→ N/A	→ 0.036	→ N/A	→ N/A	N/A
→ 30	→ 0.002	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 31	→ 0.015	→ N/A	→ N/A	→ 0.028	→ N/A	→ N/A	N/A
→ 32	→ 0.002	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 33	→ 0.031	→ N/A	→ N/A	→ 0.036	→ N/A	→ N/A	N/A
→ 34	→ 0.003	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 35	→ 0.024	→ N/A	→ N/A	→ 0.038	→ N/A	→ N/A	N/A
→ 36	→ 0.002	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 37	→ 0.035	→ N/A	→ N/A	→ 0.039	→ N/A	→ N/A	N/A
→ 38	→ 0.0032	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A
→ 39	→ 0.028	→ N/A	→ N/A	→ 0.026	→ N/A	→ N/A	N/A
→ 40	→ 0.003	→ N/A	→ N/A	→ 0.005	→ N/A	→ N/A	N/A

Note: Measured I_{ref} was applied for this test.

7.4. VOLTAGE FLUCTUATIONS AND FLICKER

LIMITS (EN61000-3-3)

Test items	Limits (EN61000-3-3)	Descriptions
P_{st}	≤ 1.0 , $T_p=10$ min	short-term flicker indicator
P_{lt}	≤ 0.65 , $T_p=2$ h	long-term flicker indicator
d_c	≤ 3.3 %	relative steady-state voltage change
d_{max}	≤ 4 % (or 6 % ^{Note(1)} , 7 % ^{Note(2)})	maximum relative voltage change:
$d_{(t)}$	≤ 3.3 %, more than 500 ms	relative voltage change characteristic

Note:

1) 6 % for equipment which is:

- a) switched manually, or
- b) switched automatically more frequently than twice per day, and also has either a delayed restart (the delay being not less than a few tens of seconds), or manual restart, after a power supply interruption.

2) 7 % for equipment which is

- a) attended whilst in use (for example: hair dryers, vacuum cleaners, kitchen equipment such as mixers, garden equipment such as lawn mowers, portable tools such as electric drills), or
- b) switched on automatically, or is intended to be switched on manually, no more than twice per day, and also has either a delayed restart (the delay being not less than a few tens of seconds) or manual restart, after a power supply interruption.

Limits (EN61000-3-11)

Test items	Limits (EN61000-3-11)	Descriptions
P_{st}	≤ 1.0 , $T_p=10$ min	short-term flicker indicator
P_{lt}	≤ 0.65 , $T_p=2$ h	long-term flicker indicator
d_c	≤ 3.3 %	relative steady-state voltage change
d_{max}	≤ 4 % (or 6 % ^{Note(1)} , 7 % ^{Note(2)})	maximum relative voltage change:
$d_{(t)}$	≤ 3.3 %, more than 500 ms	relative voltage change characteristic

Note:

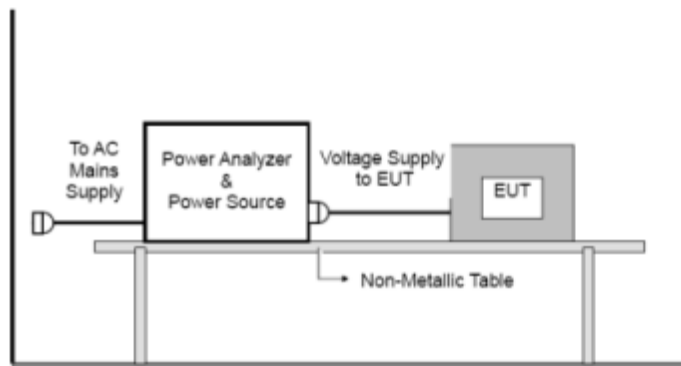
- a) 4 % without additional conditions;
- b) 6 % for equipment with:
 - manual switching, or
 - automatic switching more frequently than twice per day and having a delayed restart (the delay being not less than a few tens of seconds) or,
 - manual restart after a power supply interruption.
- c) 7 % for equipment which

- is attended whilst in use (for example: hair dryers, vacuum cleaners, kitchen equipment such as mixers, garden equipment such as lawnmowers, portable tools such as electric drills); or
- is switched on automatically, or is intended to be switched on manually, no more than twice per day and has a delayed restart (the delay being not less than a few tens of seconds) or manual restart after a power supply interruption.

TEST PROCEDURE

- 1) The EUT was placed on the top of a wooden table 0.8 meters above the ground and operated to produce the most unfavorable sequence of voltage changes under normal Condition
- 2) During the flick measurement, the measure time shall include that part of whole operation changes. The observation period for short-term flicker indicator is 10 minutes and the observation period for long-term flicker indicator is 2 hours.
- 3) Tests was performed according to the Test Condition/Assessment of Voltage Fluctuations specified in Clause 6.0/4.0 of IEC/EN 61000-3-3 depend on which standard adopted for compliance measurement.
- 4) All types of harmonic current and/or voltage fluctuation in this report are assessed by direct measurement using flicker-meter.

TEST SETUP



TEST ENVIRONMENT

Temperature	24.1°C	Relative Humidity	55.2%
Atmosphere Pressure	101kPa		

TEST DATE / ENGINEER

Test Date	March. 26, 2024	Test By	Vian
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TEST MODE

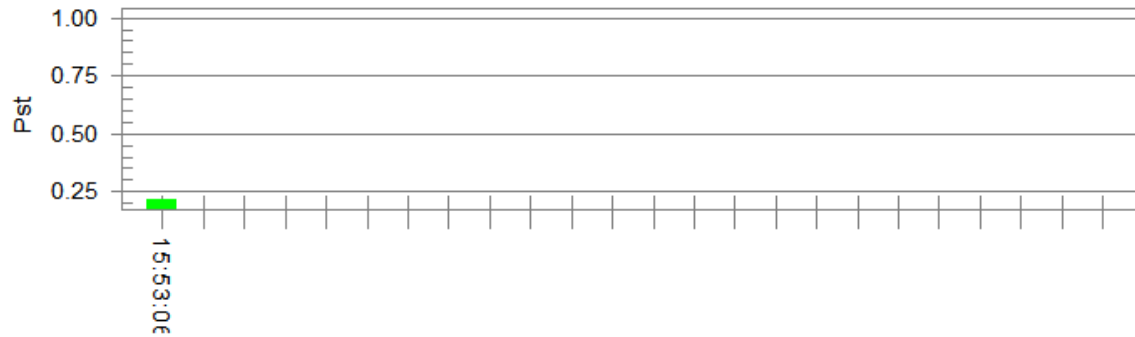
Pre-test Mode:	M01
Final Test Mode:	M01

Note: All test modes had been tested, but only the worst data recorded in the report.

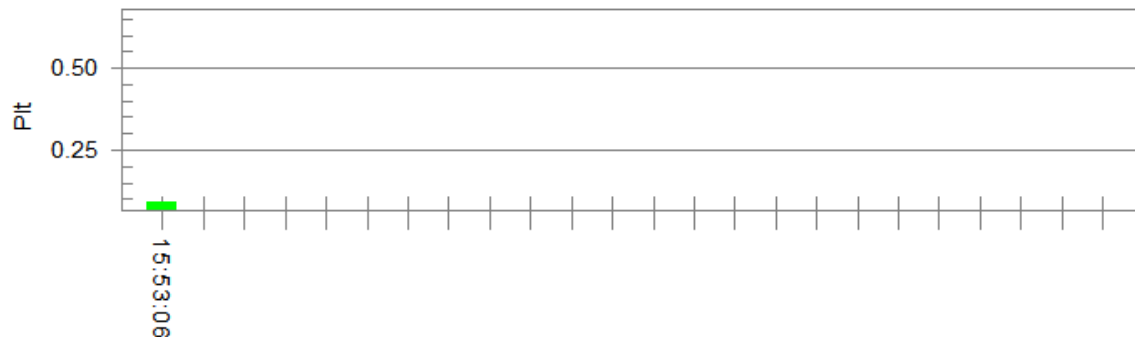
TEST RESULTS

Test Mode:	M01	Test Voltage:	AC 230V_50Hz
Test Model:	MR484000SG2-GS		

Pst and limit line



Plt and limit line



Parameter values recorded during the test:

Vrms at the end of test (Volt): → 229.64

T-max (mS):	→	0.0	→	Test limit (mS):	→	500.0	→	Pass
Highest dc (%):	→	0.00	→	Test limit (%):	→	3.30	→	Pass
Highest dmax (%):	→	0.07	→	Test limit (%):	→	4.00	→	Pass
Highest Pst (10 min. period):	→	0.216	→	Test limit:	→	1.000	→	Pass
Highest Plt (2 hr. period):	→	0.092	→	Test limit:	→	0.650	→	Pass

8. IMMUNITY TEST

8.1. PERFORMANCE CRITERIA

GENERAL PERFORMANCE CRITERIA

According to EN 55035:2017/A11:2020 standard, the general performance criteria as following:

Criteria A	The equipment shall continue to operate as intended without operator intervention. No degradation of performance, loss of function or change of operating state is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criteria B	During the application of the disturbance, degradation of performance is allowed. However, no unintended change of actual operating state or stored data is allowed to persist after the test. After the test, the equipment shall continue to operate as intended without operator intervention; no degradation of performance or loss of function is allowed, below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level (or the permissible performance loss), or recovery time, is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criteria C	Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. A reboot or re-start operation is allowed. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

According to EN300386 standard, Switching equipment specific requirements performance criteria as following:

Criterion A	During the sweep: -the established connections shall be maintained throughout testing and the transfer of information shall be within the limits of the manufacturer's specification; -loss of frame alignment or loss of synchronization is not allowed during each individual exposure (if applicable).
Criterion B	The established connections shall be maintained throughout testing except in the case of surge immunity testing at 1 kV where disconnection is allowed on the port being tested: -it shall be possible to establish a connection between two ports after the end of the transient disturbances; -it shall be possible to clear a connection in a controlled manner after the end of the transient disturbances.
Criterion C	Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls, or, in the case of switching equipment, by normal subsequent use.

PERFORMANCE CRITERIA FOR BROADCAST RECEPTION FUNCTION

The broadcast reception function shall comply with the general performance criteria given in Clause 8 and any relevant annex with the deviations defined in Table A.2.

Table A.2 – Modified test levels for performance criterion A for the broadcast reception function			
Performance criteria	Test type table clause	Group 1	Group 2
Criterion A	1.2 1.3	The disturbance level is reduced to 1 V/m for in-band frequencies.	No test requirements apply
	2.1 3.1 4.1	The disturbance level is reduced to 1 V for in-band frequencies.	

In-band is defined as the entire tuneable operating range of the selected broadcast reception function. The tuned channel $\pm 0,5$ MHz (lower edge frequency – 0,5 MHz up to the upper edge frequency + 0,5 MHz of the tuned channel) is excluded from testing.
Note: In some countries, there is a requirement to test the tuned channels. Refer to the relevant regional requirements for guidance.

PERFORMANCE CRITERIA FOR PRINT FUNCTION

Criterion A	Refer to chapter B.3.1
Criterion B	Refer to chapter B.3.2
Criterion C	Refer to chapter B.3.3

PERFORMANCE CRITERIA FOR SCAN FUNCTION

Criterion A	Refer to chapter C.3.1
Criterion B	Refer to chapter C.3.2
Criterion C	Refer to chapter C.3.3

PERFORMANCE CRITERIA FOR DISPLAY AND DISPLAY OUTPUT FUNCTION

Criterion A	Refer to chapter D.3.1 and D.3.2
Criterion B	Refer to chapter D.3.3
Criterion C	Refer to chapter D.3.4

PERFORMANCE CRITERIA FOR MUSICAL TONE GENERATING FUNCTION

Criterion A	Refer to chapter E.3.2
Criterion B	Refer to chapter E.3.3
Criterion C	Refer to chapter E.3.4

PERFORMANCE CRITERIA FOR NETWORKING FUNCTION

General requirements for network functions	
Criterion A	Refer to chapter F.3.3.1
Criterion B	Refer to chapter F.3.3.2
Criterion C	Refer to chapter F.3.3.3

Requirements for CPE containing xDSL ports	
Criterion A	Refer to chapter F.4.2
Criterion B	Refer to chapter F.4.3
Criterion C	Refer to chapter F.4.4

PERFORMANCE CRITERIA FOR AUDIO OUTPUT FUNCTION

Criterion A	Refer to chapter G.7.1
Criterion B	Refer to chapter G.7.2
Criterion C	Refer to chapter G.7.3

PERFORMANCE CRITERIA FOR TELEPHONY FUNCTION

Criterion A	Refer to chapter H.4 Table H.1
Criterion B	Refer to chapter H.4 Table H.1
Criterion C	Refer to chapter H.4 Table H.1

8.2. ELECTROSTATIC DISCHARGE

TEST SPECIFICATION

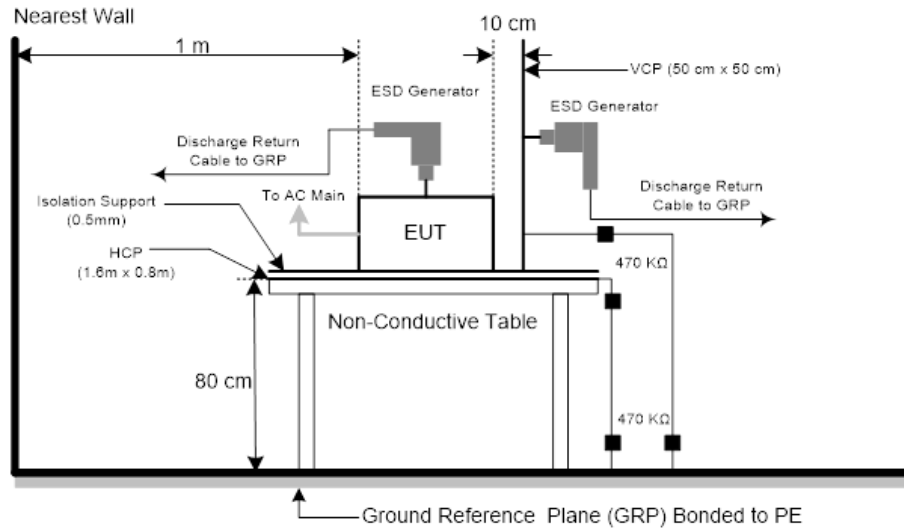
Standard:	IEC 61000-4-2(ETSI EN 300 386 and EN 55035)
Criterion Required:	Performance criteria B
Discharge Impedance:	330(1±10 %)Ω / 150(1±10 %) pF
Polarity:	Positive & Negative
Number of Discharge:	Minimum 10 times at each test point
Discharge Mode:	Single Discharge
Discharge Period:	1 second minimum
Test Level:	Air Discharge: 2 kV, 4 kV, 8 kV (Direct); Contact Discharge: 4 kV; 6 kV (Direct/Indirect) For Client requirement: Air Discharge: 10 kV (Direct); Contact Discharge: 8 kV (Direct/Indirect)

TEST PROCEDURE

The test generator necessary to perform direct and indirect application of discharges to the EUT in the following manner:

- 1) Contact discharge was applied to conductive surfaces and coupling planes of the EUT.
During the test, it was performed with single discharges. For the single discharge time between successive single discharges was at least 1 second.
- 2) Vertical Coupling Plane (VCP):
The coupling plane, of dimensions 0.5 m x 0.5 m, is placed parallel to, and positioned at a distance 0.1m from, the EUT, with the Discharge Electrode touching the coupling plane.
The four faces of the EUT will be performed with electrostatic discharge.
- 3) Horizontal Coupling Plane (HCP):
The coupling plane is placed under to the EUT. The generator shall be positioned vertically at a distance of 0.1 m from the EUT, with the Discharge Electrode touching the coupling plane.
The four faces of the EUT will be performed with electrostatic discharge.
- 4) Air discharges at insulation surfaces of the EUT.
It was at least ten single discharges with positive and negative at the same selected point.
The test shall be performed with single discharges. On each pre-selected point at least 10 single discharges (in the most sensitive polarity) shall be applied.
For air discharge testing, the test shall be applied at all test levels 2 kV, 4 kV and 8 kV. For the actual test configuration, please refer to the related Item: EUT Test Photos.

TEST SETUP



TEST ENVIRONMENT

Temperature	24.1°C	Relative Humidity	56.9%
Atmosphere Pressure	101kPa	Test Voltage	AC 230V/50Hz,

TEST DATE / ENGINEER

Test Date	April. 26, 2024	Test By	Vian
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TEST MODE

Test Mode:	M01
Test Model:	MR484000SG2-GS

TEST RESULTS

Mode	Level(kV)	Polarity	Test Point	Criteria	Result	Judgement
Contact Discharge	4	+	All Metal	B	A	Pass
Contact Discharge	4	-	All Metal	B	A	Pass
Contact Discharge	6	+	All Metal	B	A	Pass
Contact Discharge	6	-	All Metal	B	A	Pass
Horizontal Coupling	4	+	Front,rear,left,right	B	A	Pass
Horizontal Coupling	4	-	Front,rear,left,right	B	A	Pass
Vertical Coupling	4	+	Front,rear,left,right	B	A	Pass
Vertical Coupling	4	-	Front,rear,left,right	B	A	Pass
Horizontal Coupling	8	+	Front,rear,left,right	B	A	Pass
Horizontal Coupling	8	-	Front,rear,left,right	B	A	Pass
Vertical Coupling	8	+	Front,rear,left,right	B	A	Pass
Vertical Coupling	8	-	Front,rear,left,right	B	A	Pass
Air Discharge	8	+	Fan mouth panel	B	A	Pass
Air Discharge	8	-	Fan mouth panel	B	A	Pass
Air Discharge	10	+	Fan mouth panel	B	A	Pass
Air Discharge	10	-	Fan mouth panel	B	A	Pass
Contact Discharge	8	+	All Metal	B	A	Pass
Contact Discharge	8	-	All Metal	B	A	Pass

Observation:

A: No observable change.

Conclusion: The EUT met the requirements of the standard

8.3. CONTINUOUS RF ELECTROMAGNETIC FIELD DISTURBANCES

TEST SPECIFICATION

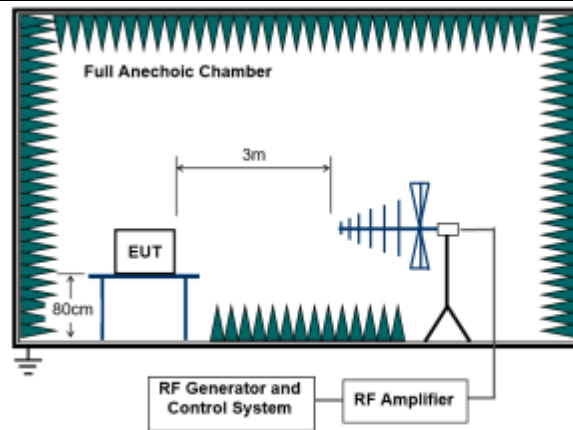
Standard:	IEC 61000-4-3(ETSI EN 300 386 and EN 55035)
Criterion Required:	Performance criteria A
Test Level:	80 MHz to 1000 MHz: 3 V/m (measured unmodulated 1 800 MHz, 2 600 MHz, 3 500 MHz, 5 000 MHz : 3 V/m (measured unmodulated 80 MHz to 690 MHz, 3V/m(rms), 1 kHz, 80%,(measured unmodulated 690 MHz to 6000 MHz, 10V/m(rms), 1 kHz, 80%,(measured unmodulated
Modulation:	The test signal shall be amplitude modulated to a depth of 80 % by a sinusoidal audio signal of 1 000 Hz.
Frequency Step:	1 % of fundamental
Dwell time:	1 seconds
Antenna Polarization:	Horizontal and vertical

TEST PROCEDURE

The test procedure was in accordance with IEC 61000-4-3.

- 1) The testing was performed in a fully anechoic chamber. The transmit antenna was located at a distance of 3 meters from the EUT.
- 2) The disturbance test signal shall be 80 % amplitude modulated by a sine wave, preferably having a frequency of 1 kHz. A frequency other than 1 kHz may be used where permitted within EN 55035 (for example Clause G.3).
- 3) 1 % step size is preferred, the frequency range can be swept incrementally with a step size not exceeding 4 % of the previous frequency with a test level of twice the value of the specified test level.
- 4) The dwell time at each frequency shall not be less than the time necessary for the EUT to be exercised and to be able to respond. However, the dwell time should not exceed 5 s at each of the frequencies during the scan.
- 5) The test was performed with the EUT exposed to both vertically and horizontally polarized fields.

TEST SETUP



Note 1: For the 1000MHz to 6000 MHz tests, the antenna will change to horn antenna.
Note 2: For the actual test configuration, please refer to Appendix I: Photographs of the Test Configuration.

TEST ENVIRONMENT

Temperature	24.3°C	Relative Humidity	56.5%
Atmosphere Pressure	101kPa	Test Voltage	AC 230V/50Hz, AC 110V/60Hz

TEST DATE / ENGINEER

Test Date	April. 26, 2024	Test By	Vian
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TEST MODE

Test Mode:	M01
Test Model:	MR484000SG2-GS

TEST RESULTS

Freq.Range (MHz)	Position (Face)	Polarity (H or V)	Field Strength (V/m) (unmodulated,r.m.s)	Criterion	Result	Judgment
80-1000, 1800, 2600, 3500, 5000,	0°	H&V	3 V/m	A	A	Pass
80-1000, 1800, 2600, 3500, 5000,	90°	H&V	3 V/m	A	A	Pass
80-1000, 1800, 2600, 3500, 5000,	180°	H&V	3 V/m	A	A	Pass
80-1000, 1800, 2600, 3500, 5000,	270°	H&V	3 V/m	A	A	Pass
690-6000	0°	H&V	10 V/m	A	A	Pass
690-6000	90°	H&V	10 V/m	A	A	Pass
690-6000	180°	H&V	10 V/m	A	A	Pass
690-6000	270°	H&V	10 V/m	A	A	Pass
Observation: A: No observable change. Conclusion: The EUT met the requirements of the standard						

8.4. ELECTRICAL FAST TRANSIENTS BURST (AC MAINS POWER PORTS)

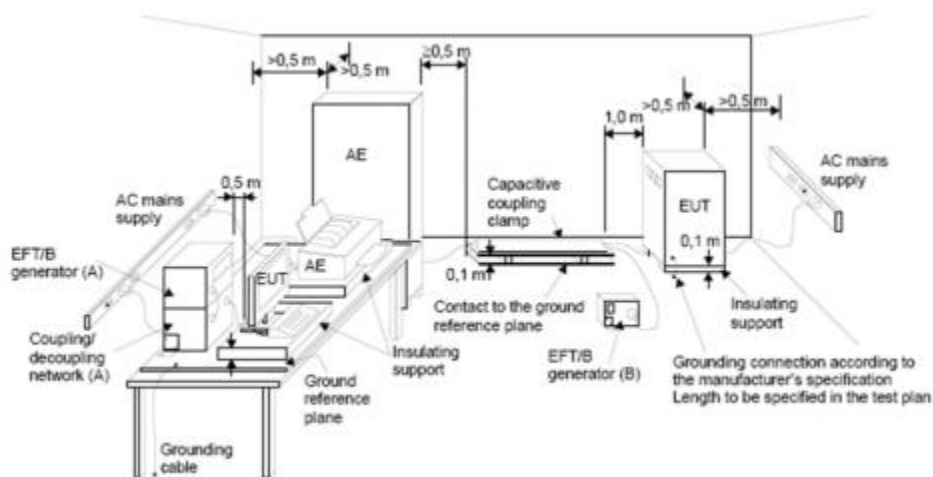
TEST SPECIFICATION

Standard:	IEC 61000-4-4(ETSI EN 300 386 and EN 55035)
Criterion Required:	Performance criteria B
Polarity:	Positive & Negative
Test Level and Repetition Frequency:	The test level for shall be 0.5kV,1 kV open circuit voltage at a repetition rate of 5 kHz as given EN 61000-4-4. For Client requirement: AC mains power ports: 4.0kV(peak)
Impulse Wave shape:	5/50 Tr/Th ns
Burst Duration:	15 ms
Burst Period:	300 ms
Test Duration:	1 Minute

TEST PROCEDURE

- 1) Both positive and negative polarity discharges were applied.
- 2) The duration time of each test sequential was 1 minute.
- 3) The transient/burst waveform was in accordance with IEC 61000-4-4, 5/50ns.
- 4) Multi conductor cables shall be tested as a single cable. Cables shall not be split or divided into groups of conductors for this test.

TEST SETUP



TEST ENVIRONMENT

Temperature	24.3℃	Relative Humidity	56.4%
Atmosphere Pressure	101kPa	Test Voltage	AC 230V/50Hz, AC 110V/60Hz

TEST DATE / ENGINEER

Test Date	March. 26, 2024	Test By	Vian
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TEST MODE

Test Mode:	M01
Test Model:	MR484000SG2-GS

TEST RESULTS

Coupling Line	Test Levels(kV)	Polarity	Criteria	Results	Judgement
L	1, 4	+	B	A	Pass
L	1, 4	-	B	A	Pass
N	1, 4	+	B	A	Pass
N	1, 4	-	B	A	Pass
PE	1, 4	+	B	A	Pass
PE	1, 4	-	B	A	Pass
L-N	1, 4	+	B	A	Pass
L-N	1, 4	-	B	A	Pass
L-PE	1, 4	+	B	A	Pass
L-PE	1, 4	-	B	A	Pass
N-PE	1, 4	+	B	A	Pass
N-PE	1, 4	-	B	A	Pass
L-N-PE	1, 4	+	B	A	Pass
L-N-PE	1, 4	-	B	A	Pass
DC network power ports	0.5	+	B	/	/
DC network power ports	0.5	-	B	/	/

Observation:

A: No observable change.

Conclusion: The EUT met the requirements of the standard

8.5. SURGES (AC MAINS POWER PORTS)

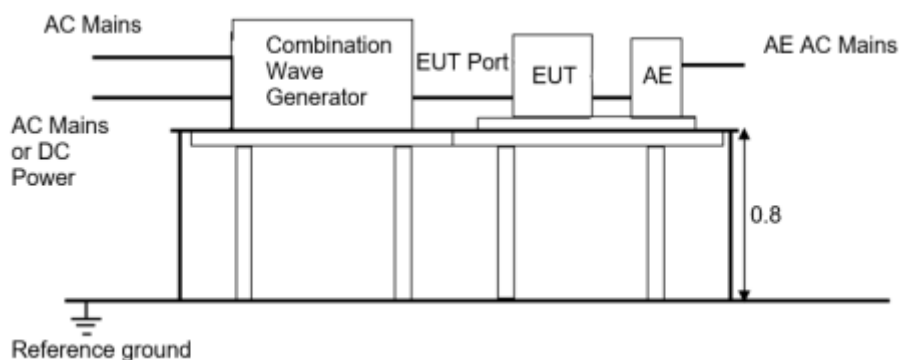
TEST SPECIFICATION

Standard:	IEC 61000-4-5 (ETSI EN 300 386 and EN 55035)
Criterion Required:	Performance criteria B
Wave Shape:	Tr/Th 1.2/50µs or 10/700 us
Test Level:	1 kV (Line to Line for AC mains power ports) 2 kV (Line to Ground for AC mains power ports) 1 kV (Lines to Ground for Analogue/Digital data ports) 0.5 kV (shield to ground for coaxial/shielded cable on Analogue/Digital data ports) 0.5 kV (each individual line to reference ground for DC network power ports) For Client requirement: Line to line: 2 kV, line to ground: 4 kV
Polarity:	Positive & Negative
Interval:	60s between each surge
No. of Surges:	0°/90°/180°/270° for AC port

TEST PROCEDURE

- 1) The EUT and the auxiliary equipment were placed on a table of 0.8m heights above a metal ground reference plane. The size of ground plane is greater than 1m×1m and project beyond the EUT by at least 0.1m on all sides. The ground plane is connected to the protective earth. The length of power cord between the coupling device and the EUT was less than 2 meters (provided by the manufacturer).
- 2) The EUT was connected to the power mains through a coupling device that directly couples the surge interference signal. The surge noise was applied synchronized to the voltage phase at the zero crossing and the peak value of the AC voltage wave (positive and negative).
- 3) The surges were applied line to line and line(s) to earth. When testing line to earth the test voltage was applied successively between each of the lines and earth. Steps up to the test level specified increased the test voltage. All lower levels including the selected test level were tested. The polarity of each surge level included positive and negative test pulses.

TEST SETUP



TEST ENVIRONMENT

Temperature	24.6°C	Relative Humidity	57.9%
Atmosphere Pressure	101kPa	Test Voltage	AC 230V/50Hz, AC 110V/60Hz

TEST DATE / ENGINEER

Test Date	April. 26, 2024	Test By	Vian
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TEST MODE

Test Mode:	M01
Test Model:	MR484000SG2-GS

TEST RESULTS

Wave Form EUT Ports Tested	1.2/50(8/20)Ti/Th us						Criteria	Results	Judgement
	Polarity	Phase	Voltage						
			0.5kV	1kV	2kV	4kV			
L - N	+	0°	/	A	A	/	B	A	Pass
	-		/	A	A	/			
	+	90°	/	A	A	/			
	-		/	A	A	/			
	+	180°	/	A	A	/			
	-		/	A	A	/			
	+	270°	/	A	A	/			
	-		/	A	A	/			
L - PE	+	0°	/	/	A	A	B	A	Pass
	-		/	/	A	A			
	+	90°	/	/	A	A			
	-		/	/	A	A			
	+	180°	/	/	A	A			
	-		/	/	A	A			
	+	270°	/	/	A	A			
	-		/	/	A	A			
N - PE	+	0°	/	/	A	A	B	A	Pass
	-		/	/	A	A			
	+	90°	/	/	A	A			
	-		/	/	A	A			
	+	180°	/	/	A	A			
	-		/	/	A	A			
	+	270°	/	/	A	A			
	-		/	/	A	A			
Signal Line	+	/	/	/	/	/	B	/	/
	-		/	/	/	/			

Note:

A: No degradation in the performance of the EUT was observed.

8.6. CONTINUOUS INDUCED RF DISTURBANCES (AC MAINS POWER PORTS)

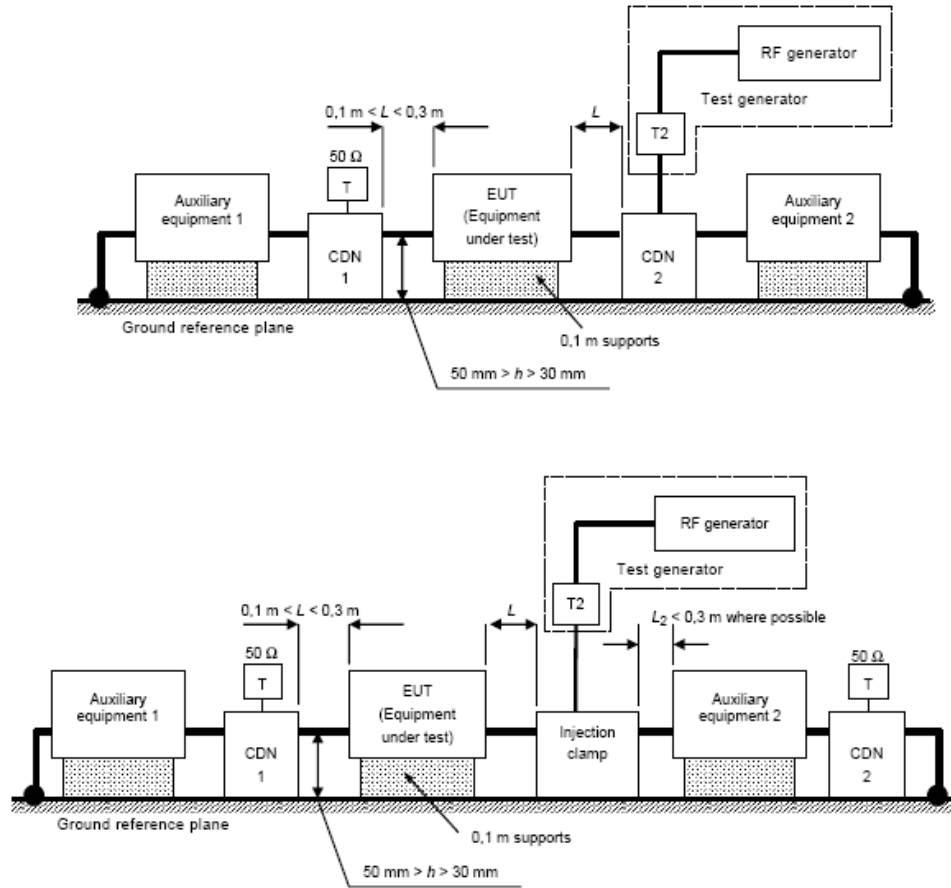
TEST SPECIFICATION

Standard:	IEC 61000-4-6(ETSI EN 300 386 and EN 55035)
Criterion Required:	Performance criteria A
Test Level:	0.15 MHz to 10 MHz: 3 V (r.m.s) 10 MHz to 30 MHz: 3 to 1 V (r.m.s) 30 MHz to 80 MHz: 1 V (r.m.s) 0.15 MHz to 80 MHz: 3 V (r.m.s) For Client requirement: 0.15 MHz - 80 MHz - 10V (r.m.s.)
Modulation:	80%, 1kHz Amplitude Modulation
Step Size:	1% increment
Dwell Time:	1s

TEST PROCEDURE

- 1) The EUT shall be tested within its intended operating and climatic conditions.
- 2) The test shall be performed with the test generator connected to each of the coupling and decoupling devices in turn, while the other non-excited RF input ports of the coupling devices are terminated by a 50-ohm load resistor.
- 3) The frequency range is swept from 150 kHz to 80 MHz, using the signal level established during the setting process and with a disturbance signal of 80 % amplitude. The signal is modulated with a 1 kHz sine wave, pausing to adjust the RF signal level or the switch coupling devices as necessary. The sweep rate shall not exceed 1.5×10^{-3} decades/s. The step size shall not exceed 1 % of the start and thereafter 1 % of the preceding frequency value where the frequency is swept incrementally.
- 4) The dwell time at each frequency shall not be less than the time necessary for the EUT to be exercised, and able to respond. Sensitive frequencies such as clock frequencies and harmonics or frequencies of dominant interest, shall be analyzed separately.
- 5) Attempts should be made to fully exercise the EUT during test, and to fully interrogate all exercise modes selected for susceptibility.

TEST SETUP



TEST ENVIRONMENT

Temperature	24.6°C	Relative Humidity	57.9%
Atmosphere Pressure	101kPa	Test Voltage	AC 230V

TEST DATE / ENGINEER

Test Date	April. 26, 2024	Test By	Vian
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TEST MODE

Test Mode:	M01
Test Model:	MR484000SG2-GS

TEST RESULTS

For ETSI EN 300 386

Test Ports (Mode)	Freq.Range (MHz)	Field Strength (unmodulated,r.m.s)	Criteria	Results	Judgment
AC mains power ports	0.15---80	3 V	A	A	Pass
AC mains power ports	0.15---80	10 V	A	A	Pass
DC network power ports	0.15---10	3 V	A	/	/
DC network power ports	10---30	3 V to 1V	A	/	/
DC network power ports	30---80	1 V	A	/	/
Analogue/digital data ports	0.15---10	3 V	A	/	/
Analogue/digital data ports	10---30	3 V to 1V	A	/	/
Analogue/digital data ports	30---80	1 V	A	/	/

Observation:

A: No observable change.

Conclusion: The EUT met the requirements of the standard

For EN 55035

Test Ports (Mode)	Freq.Range (MHz)	Field Strength (unmodulated,r.m.s)	Criteria	Results	Judgment
AC mains power ports	0.15---10	3 V	A	A	Pass
AC mains power ports	10---30	3 V to 1V	A	A	Pass
AC mains power ports	30---80	1 V	A	A	Pass
DC network power ports	0.15---10	3 V	A	/	/
DC network power ports	10---30	3 V to 1V	A	/	/
DC network power ports	30---80	1 V	A	/	/
Analogue/digital data ports	0.15---10	3 V	A	/	/
Analogue/digital data ports	10---30	3 V to 1V	A	/	/
Analogue/digital data ports	30---80	1 V	A	/	/

Observation:

A: No observable change.

Conclusion: The EUT met the requirements of the standard

8.7. ELECTRICAL FAST TRANSIENTS BURST (AC MAINS POWER PORTS)

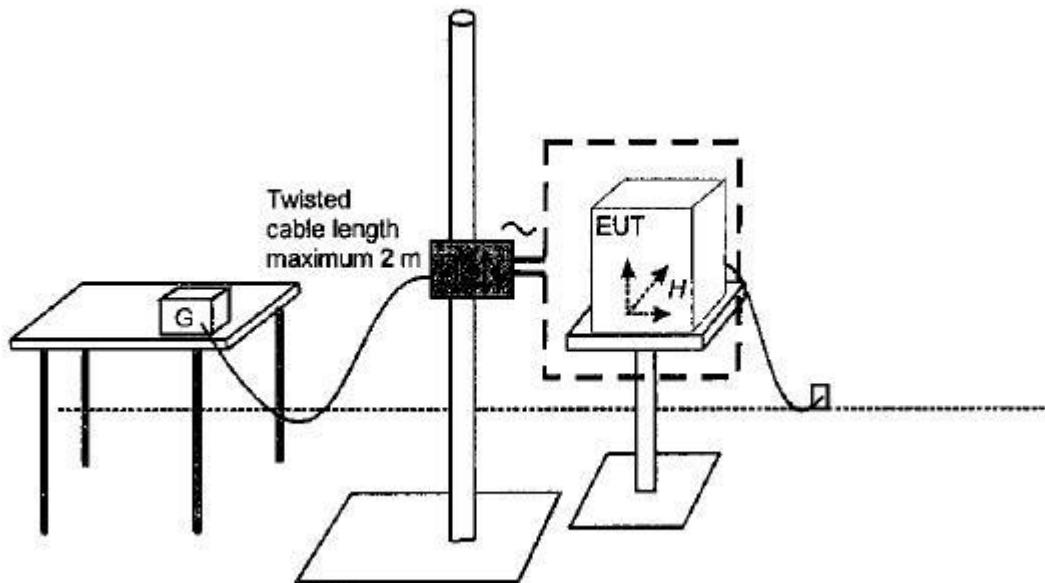
TEST SPECIFICATION

Standard:	IEC 61000-4-8 (ETSI EN 300 386 and EN 55035)
Criterion Required:	Performance criteria A
Frequency:	50Hz / 60Hz
Field Strength:	1A/m. For Client requirement: 30A/m.

TEST PROCEDURE

- Both positive and negative polarity discharges were applied.
- The EUT and its load are placed on a table which is 0.8 meter above a metal ground plane measured at least 1m*1m min. The test magnetic field shall be placed at central of the induction coil.
- The test magnetic Field shall be applied 10 minutes by the immersion method to the EUT. And the induction coil shall be rotated by 90° in order to expose the EUT to the test field with different orientation (X,Y,Z Orientations)

TEST SETUP



TEST ENVIRONMENT

Temperature	24.3°C	Relative Humidity	56.4%
Atmosphere Pressure	101kPa	Test Voltage	AC 230V/50Hz,

TEST DATE / ENGINEER

Test Date	April. 26, 2024	Test By	Vian
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TEST MODE

Test Mode:	M01
Test Model:	MR484000SG2-GS

TEST RESULTS

Test Ports (Mode)	Frequency. (Hz)	Field Strength	Required Performance Criteria	Performance Criteria Complied To	Judgment
X Orientation	50	30A/m	A	A	PASS
Y Orientation	50		A	A	PASS
Z Orientation	50		A	A	PASS

Observation:

A: No observable change.

Conclusion: The EUT met the requirements of the standard

8.8. VOLTAGE DIPS AND INTERRUPTIONS (AC MAINS POWER PORTS)

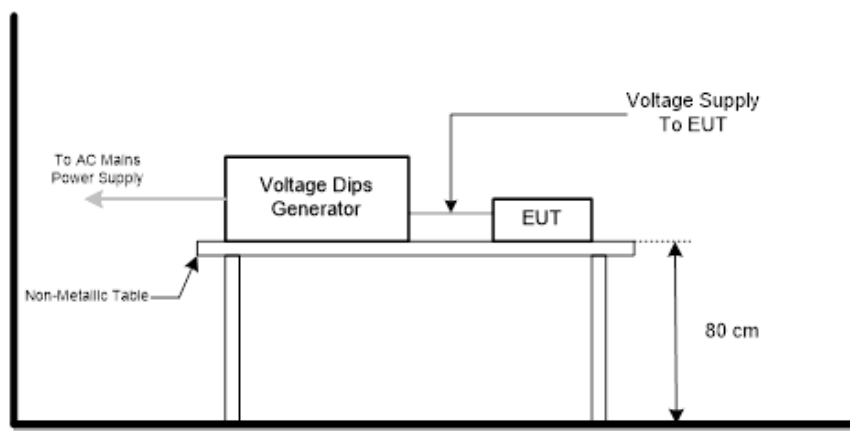
TEST SPECIFICATION

Standard:	IEC 61000-4-11 (ETSI EN 300 386 and EN 55035)
Criterion Required:	Voltage dips: performance criteria B or C; Interruptions: performance criteria C
Test Port:	AC mains power port
Test Level:	For EN55035: >95 % reduction: 0.5 period >30 % reduction: 25 period for 50Hz/ 30 period for 60Hz >95 % reduction: 250 period for 50Hz/ 300 period for 60Hz For ETSI EN 300 386: Residual voltage: 0% 1 period Voltage Dips Residual voltage:70% 25period(50Hz), Voltage Dips Residual voltage:0% 250period(50Hz), Voltage Interruptions)
Voltage Dips:	0 % Residual; 70 % Residual
Voltage Interruptions:	0 % Residual
No. of Dips / Interruptions:	3 per Level
Interval between Event:	Minimum 10 seconds
Phase Angle:	0°

TEST PROCEDURE

- 1) The power cord was used as supplied by the manufacturer. The EUT was connected to the line output of the Voltage Dips and Interruption Generator.
- 2) Voltage reductions occur at 0degree crossover point of the voltage waveform. The performance of the EUT was checked after the voltage dip or interruption.

TEST SETUP



TEST ENVIRONMENT

Temperature	24.6°C	Relative Humidity	57.9%
Atmosphere Pressure	101kPa	Test Voltage	AC 240V/50Hz, AC 100V/60Hz

TEST DATE / ENGINEER

Test Date	April. 26, 2024	Test By	Vian
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TEST MODE

Test Mode:	M01
Test Model:	MR484000SG2-GS

TEST RESULTS

Voltage(AC)	Interruption & Dips	Durations(T)	Voltage Reduction	Perform Criteria	Results	Judgment
240V 50Hz	Voltage dips	0.5 Cycles	>95%	B	A	Pass
240V 50Hz	Voltage dips	25 Cycles	30%	C	A	Pass
240V 50Hz	Voltage interruptions	250 Cycles	>95%	C	B	Pass
100V 60Hz	Voltage dips	0.5 Cycles	>95%	B	A	Pass
100V 60Hz	Voltage dips	30 Cycles	30%	C	A	Pass
100V 60Hz	Voltage interruptions	300 Cycles	>95%	C	B	Pass

Observation:

A: No observable change.

B: The device is powered off during the test, automatic recovery after test.

Conclusion: The EUT met the requirements of the standard

Interruption & Dips	Durations	Voltage Residual	Perform Criteria	Results	Judgment
Voltage dips	1 cycle	0 %	B	A	Pass
	25 cycles	70 %	C	A	
Voltage interruptions	250 cycles	0 %	C	B	

Note:

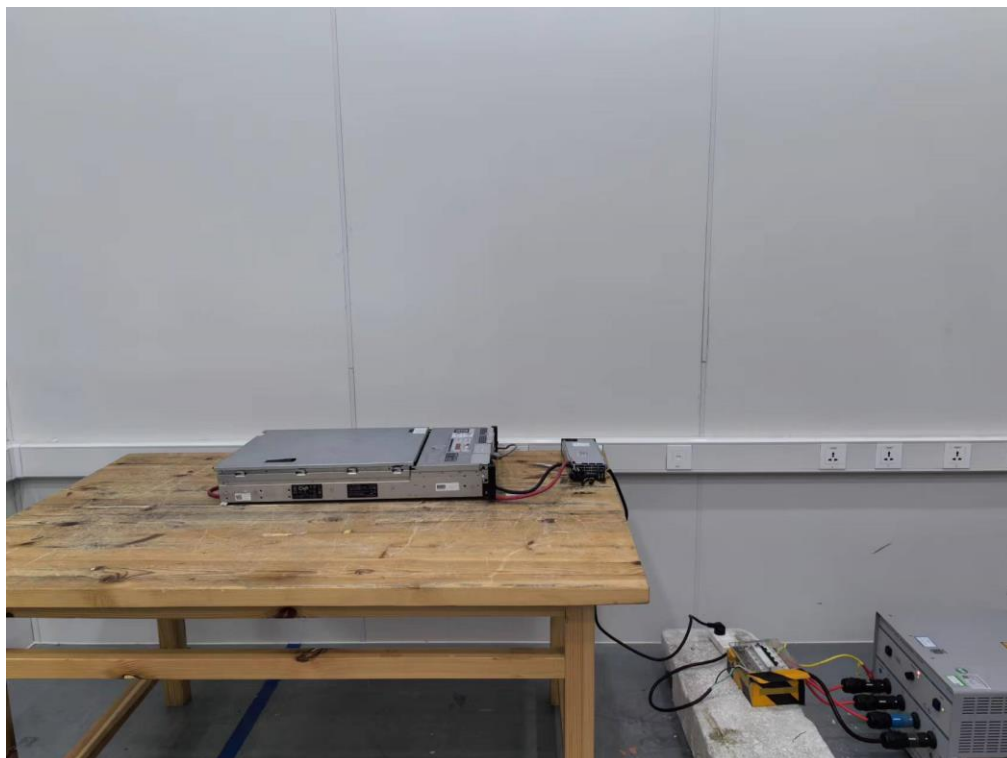
A: No degradation in the performance of the EUT was observed.

B: The device is powered off during the test, automatic recovery after test.

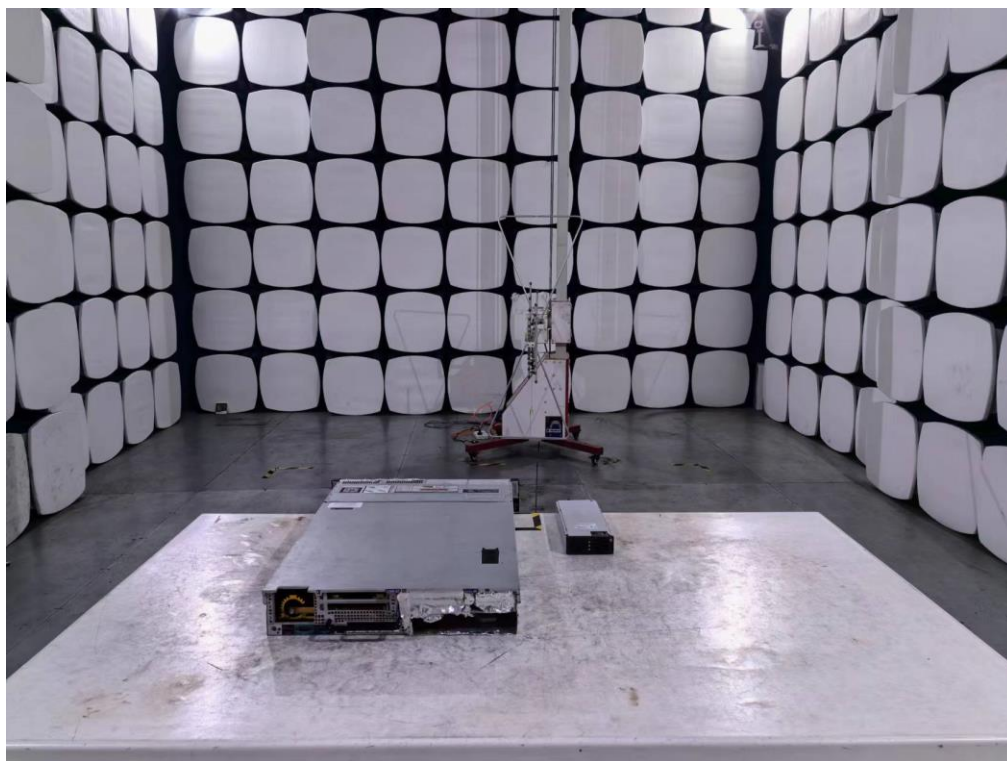
C: The device was restarted during the test.

APPENDIX: PHOTOGRAPHS OF TEST CONFIGURATION

Conducted emissions (AC mains power ports)



Radiated emissions below 1GHz



Radiated emissions below 1GHz



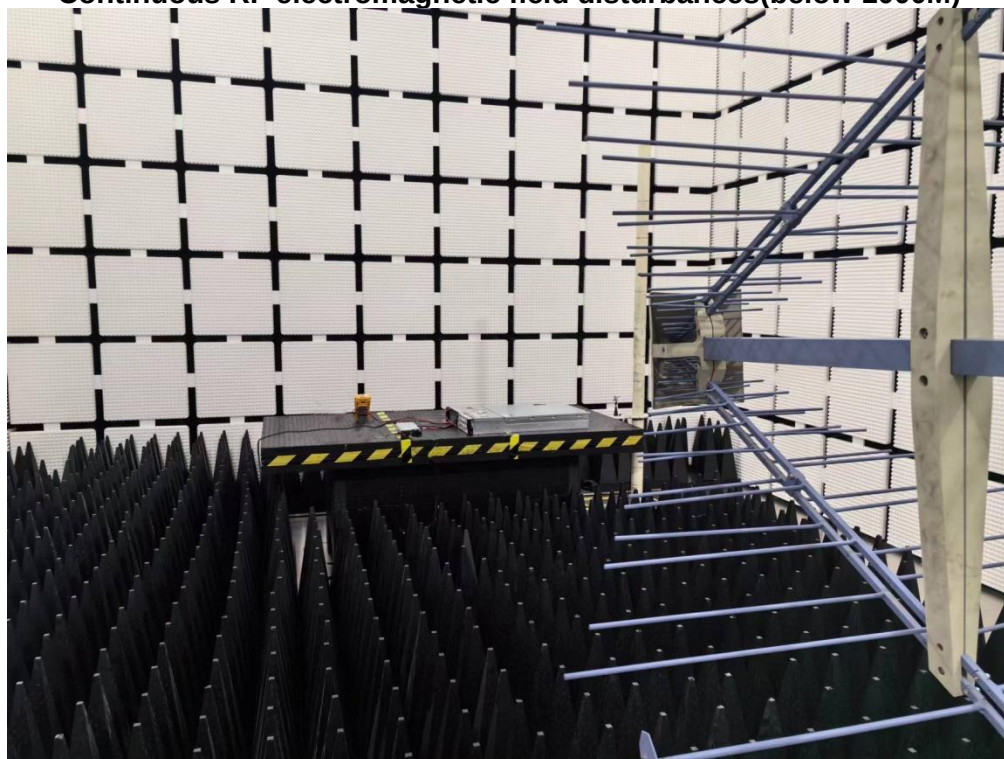
Voltage fluctuations and flicker



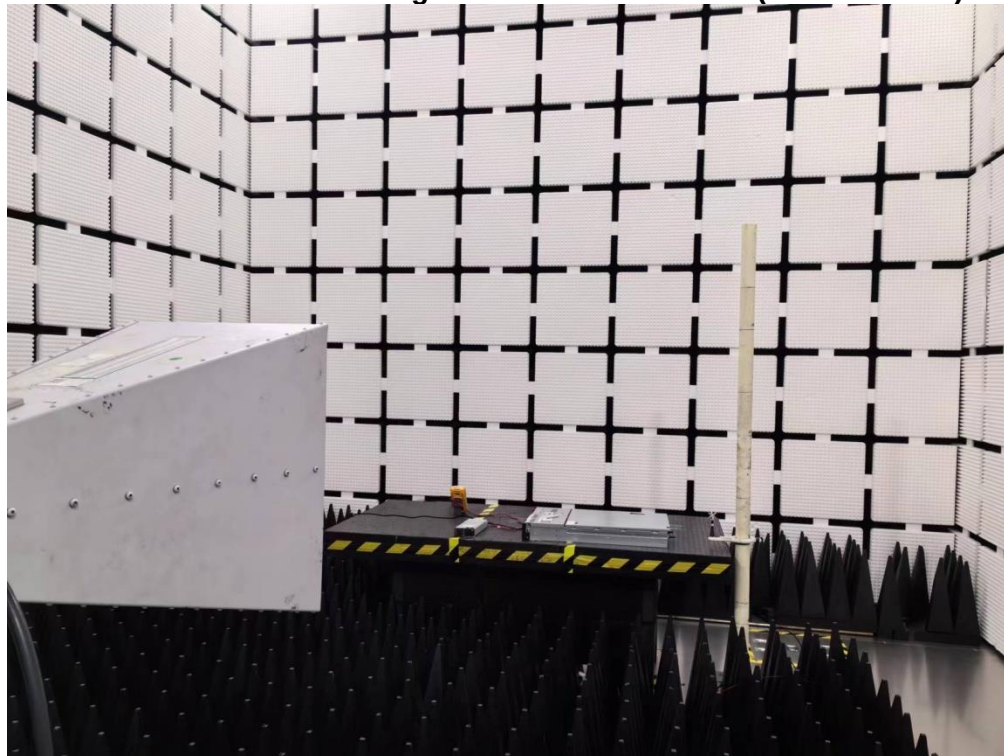
Electrostatic Discharge



Continuous RF electromagnetic field disturbances(below 1000M)



Continuous RF electromagnetic field disturbances(Above 1000M)



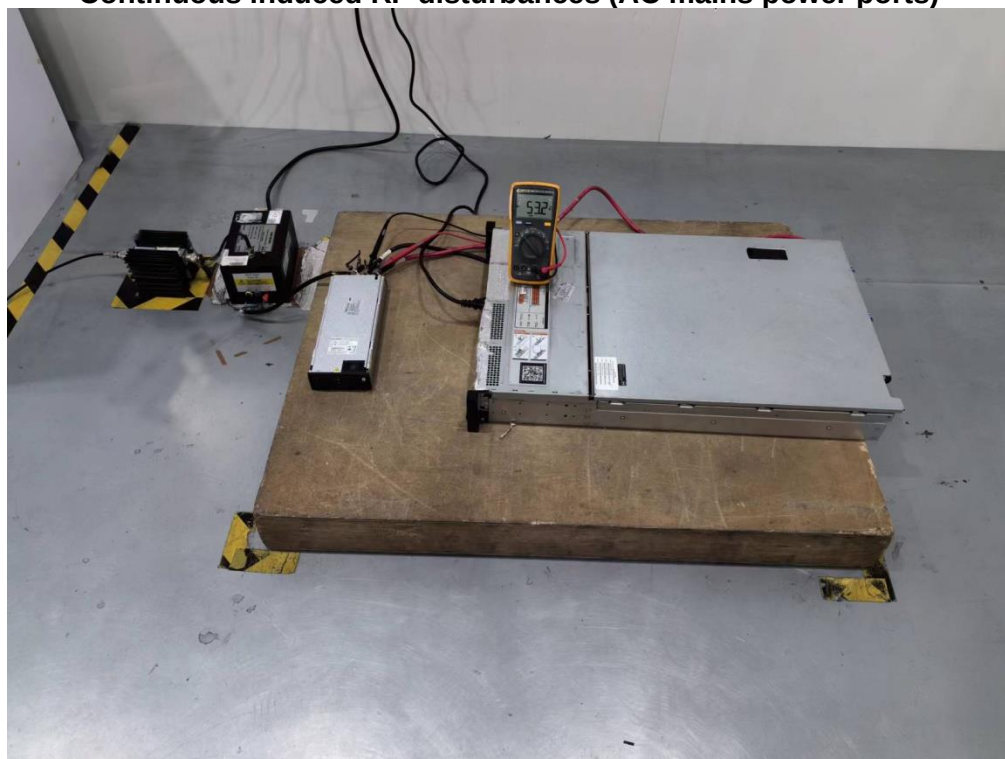
Electrical fast transients burst (AC mains power ports)



Surges (AC mains power ports)



Continuous induced RF disturbances (AC mains power ports)



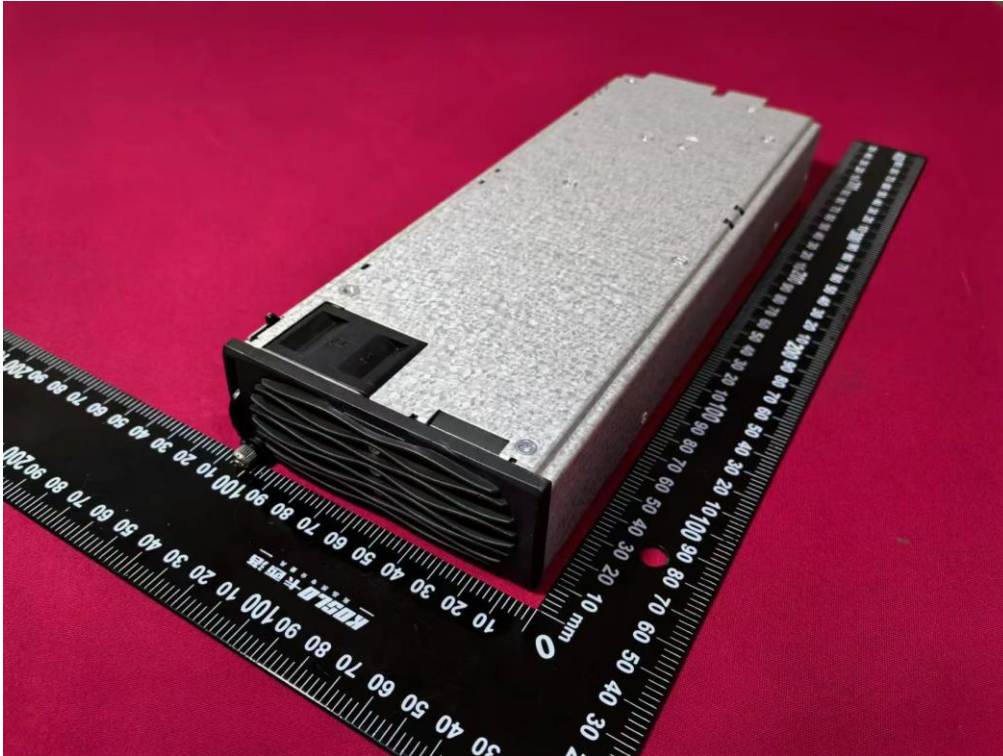
Voltage dips and interruptions (AC mains power ports)



APPENDIX: PHOTOGRAPHS OF THE EUT

External

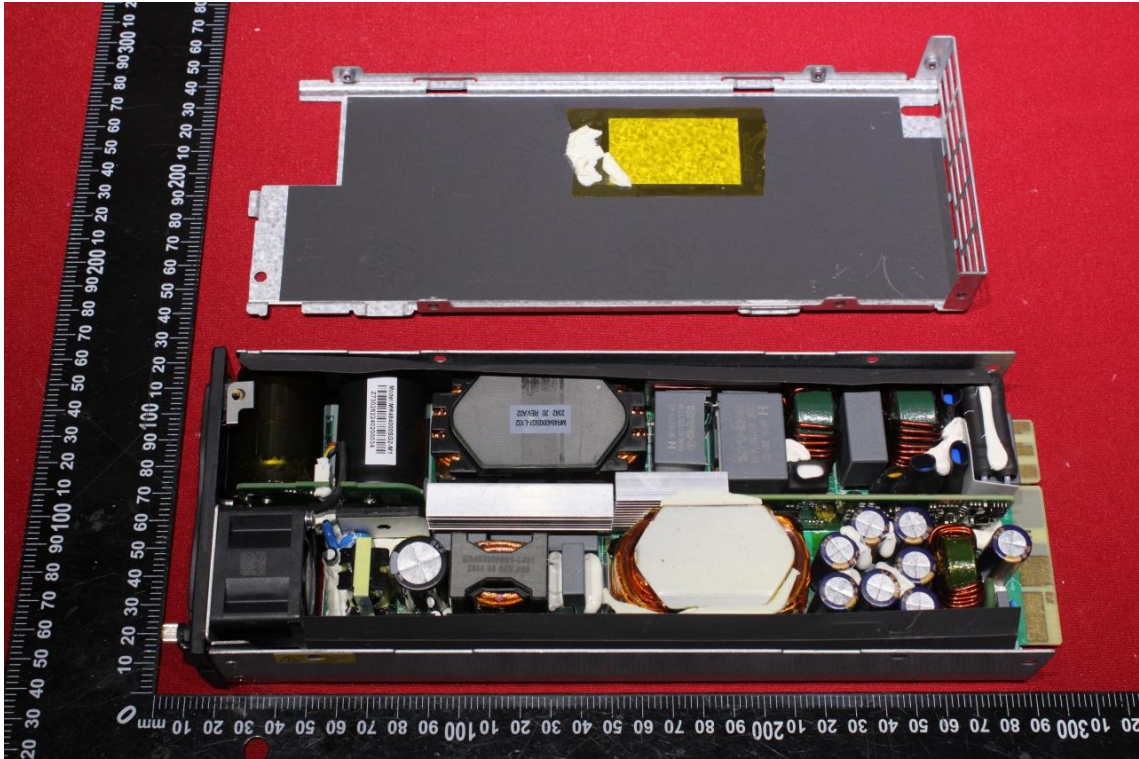
Overall View for model MR484000SG2XXXXXX



Overall View for model MR484000SG2XXXXXX



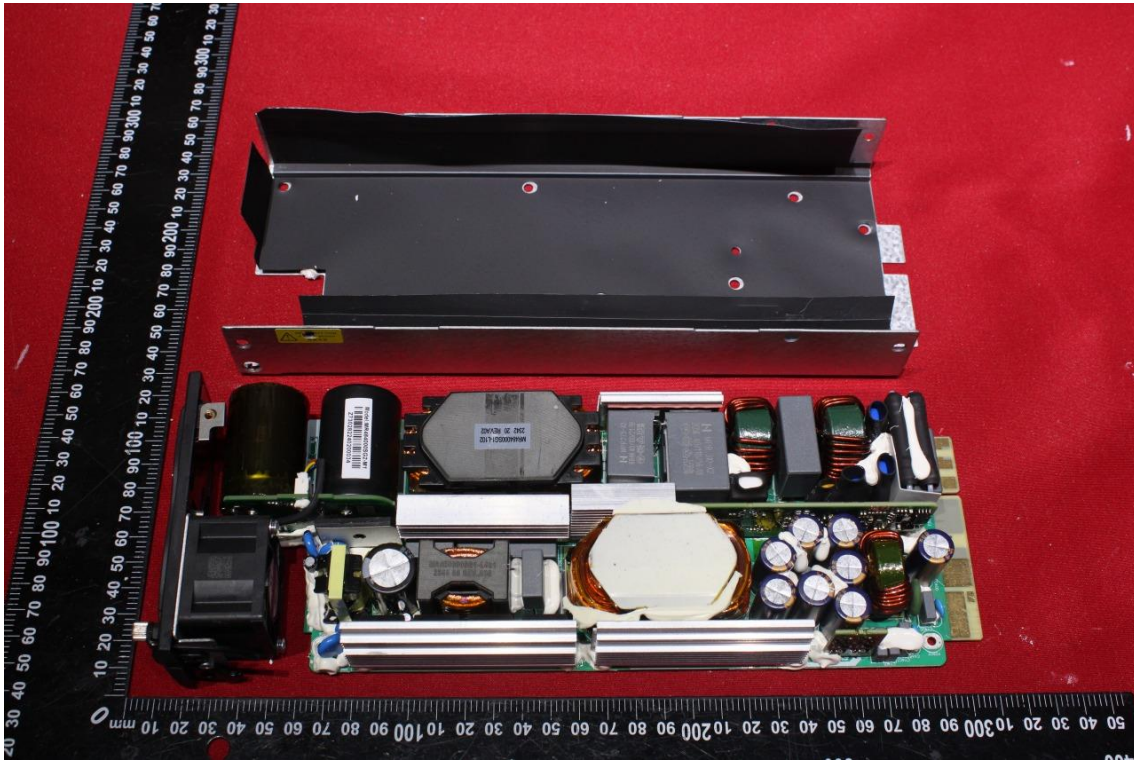
Internal view for model MR484000SG2XXXXXX



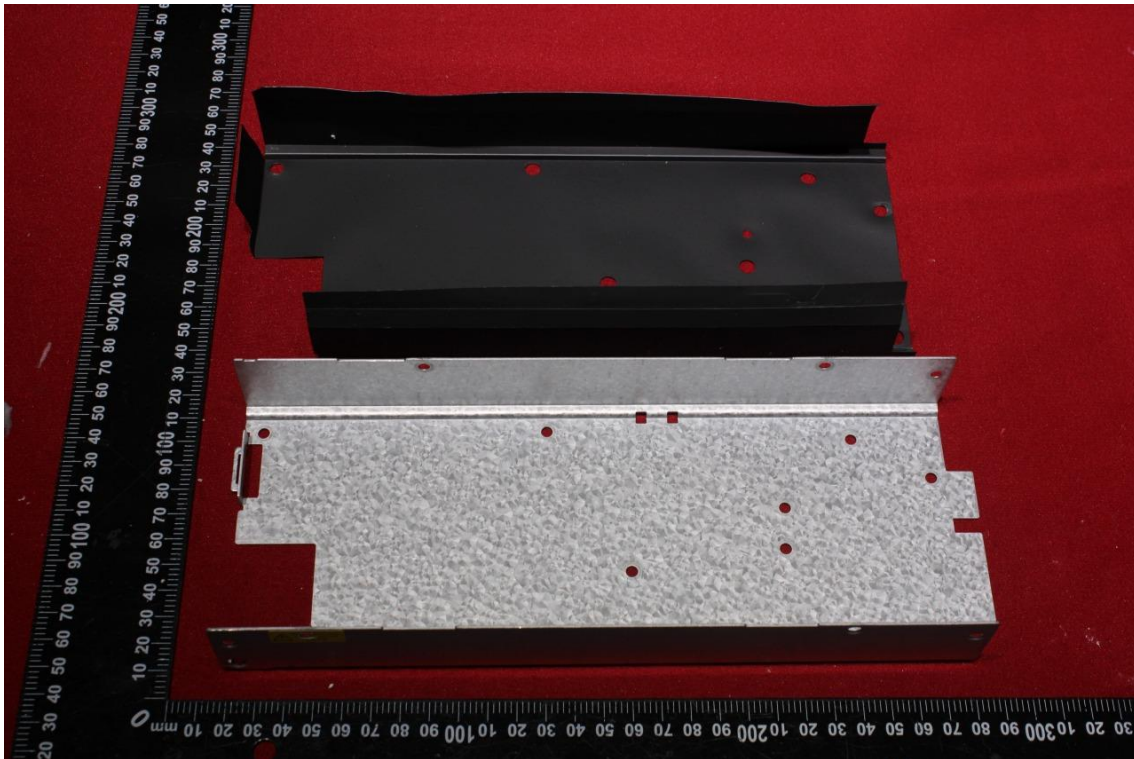
Internal view for model MR484000SG2XXXXXX



Internal view for model MR484000SG2XXXXXX



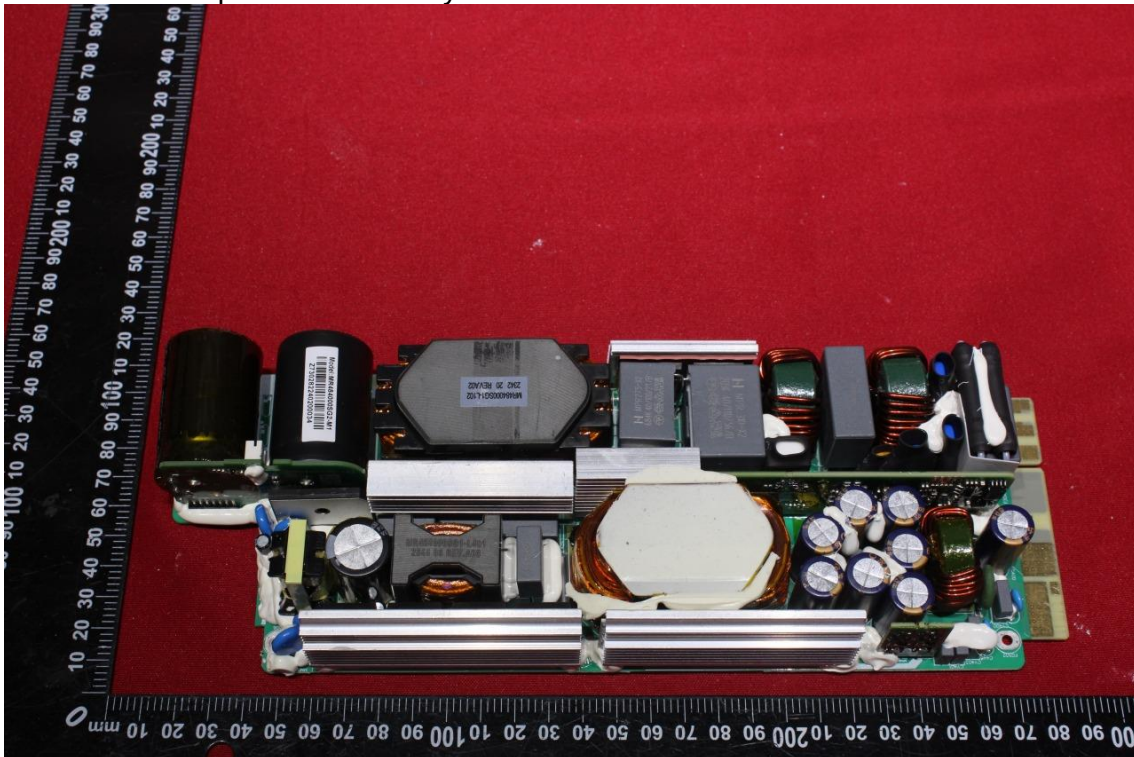
Internal view for model MR484000SG2XXXXXX



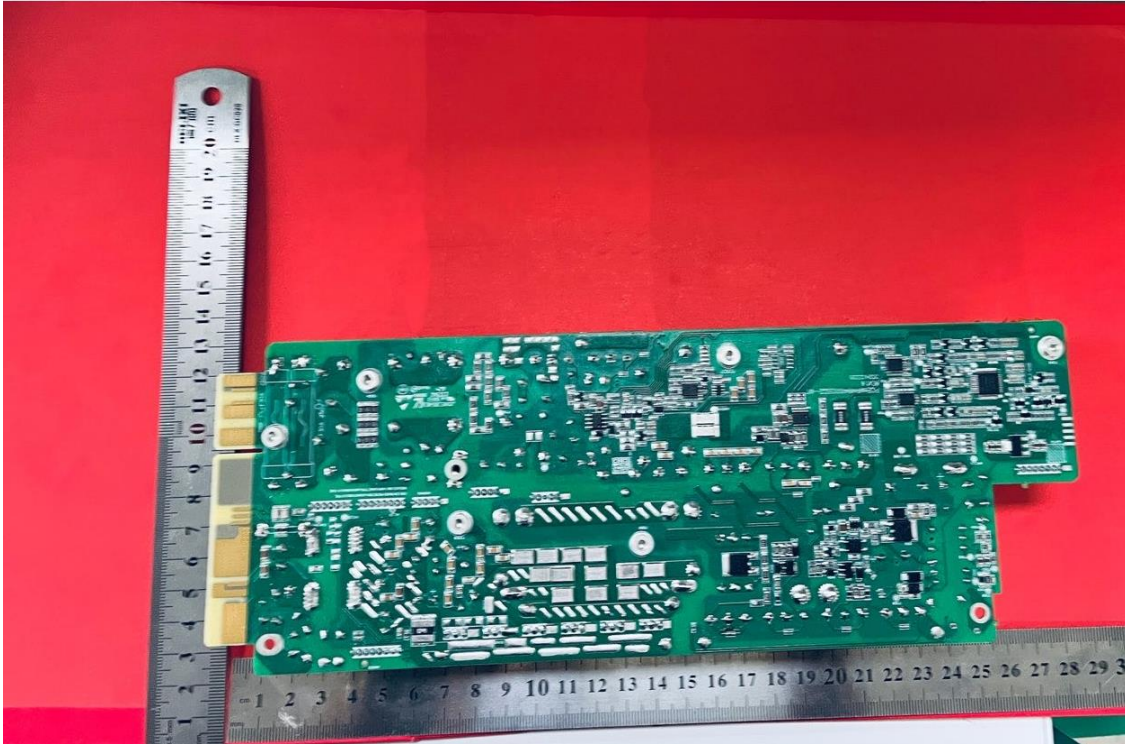
Top view for PCB layout for model MR484000SG2XXXXXX



Top view for PCB layout for model MR484000SG2XXXXXX



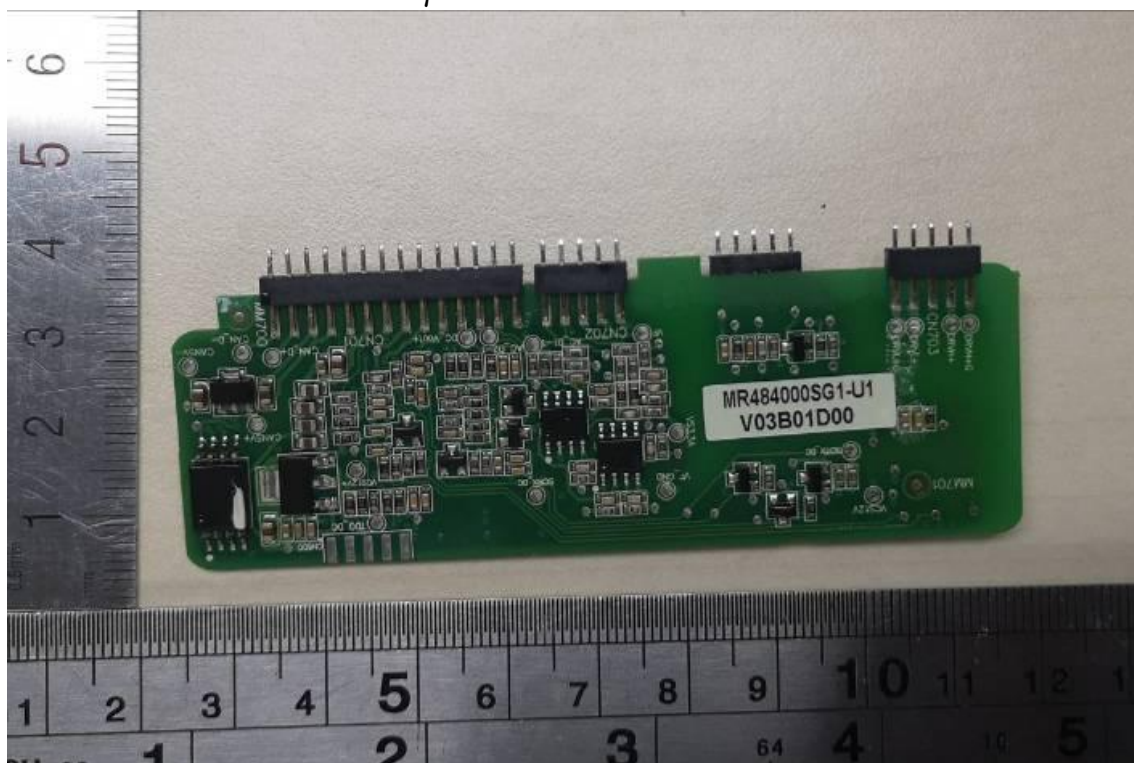
Bottom view for PCB layout for model MR484000SG2XXXXXX



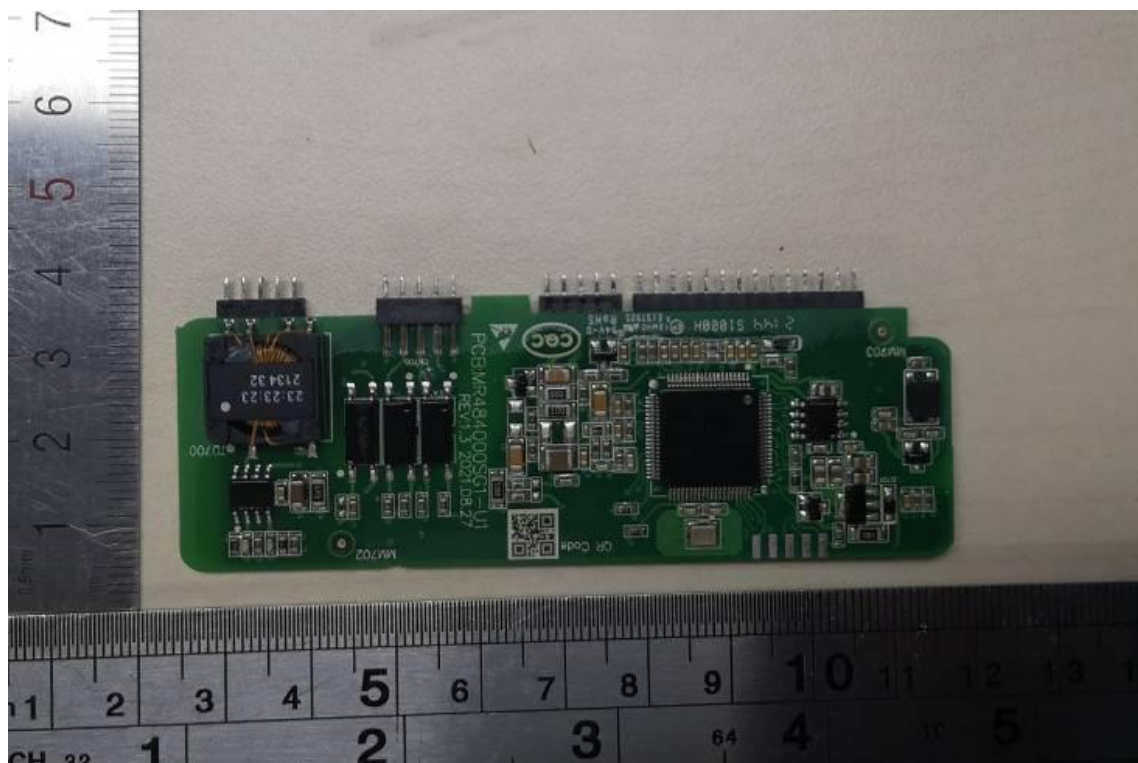
DC Fan view



Top view for main board



Bottom view for main board



Top view for main board



Bottom view for main board



END OF REPORT